

Accounting for Material Plasticity to Reduce Displacement-Controlled Moments in Piping Stress Analysis: The RCC-MRx Abatement Coefficient

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Abstract

Displacement-controlled (secondary) loads in nuclear piping systems are self-limiting and are therefore assessed less severely than primary loads. In RCC-MRx, this distinction is formalised through the *elastic follow-up factor* r_M and the associated *abatement coefficient* g , which reduce the displacement-dependent moment when material plasticity is taken into account. This paper consolidates the historical and theoretical background of the method — from beam theory, elbow ovalisation and Markl’s fatigue framework to Roche’s elastic follow-up construction — and benchmarks the RCC-MRx analytical procedure (RB 3643 and RB 3651) against a full elasto-plastic shell finite-element analysis of a simple elbow-and-straight-pipe test case. The elastic benchmark is first calibrated between a beam-based piping code (Aspect Nuclear Pipe Stress) and an MSC Marc shell model through the elbow flexibility factor k . The non-linear benchmark then uses a Ramberg–Osgood constitutive law representative of 316L(N) stainless steel at 550 °C. The RCC-MRx result ($r_M = 2.638$, $g = 0.3237$) is compared with the FE-based interpretation of the critical-section response ($g \approx 0.436$, $\sigma_{VM} = 86.5$ MPa). The apparent discrepancy is shown to arise from both the code-mandated conversion to primary-equivalent stress and the mesh sensitivity of the shell reference solution. Additional validation is provided by wall-thickness variants and by a cantilever case with a closed-form solution for r_M . The paper concludes by identifying the main modelling limitations and the implications for future pressurised benchmarks.

Keywords: RCC-MRx; elastic follow-up; abatement coefficient; displacement-controlled loading; piping stress analysis; Ramberg–Osgood law.

1 Introduction

Piping stress analysis distinguishes between *primary* loads (pressure, weight, seismic inertia, etc.), which are not self-limiting and can lead to unbounded deformation if the material yields, and *secondary* (displacement-controlled) loads, such as thermal expansion, which are self-limiting: once the material yields locally, the associated stresses relax and the deformation compatibility is preserved elastically or elasto-plastically by the remainder of the structure. Design codes therefore allow secondary stresses to be evaluated with more permissive criteria than primary stresses, provided the self-limiting character of the load is correctly represented.

The RCC-MRx code for high-temperature nuclear components formalises this distinction through the concept of the *abatement coefficient* g (RB 3651) [8], which reduces the displacement-dependent moment used in the primary-type damage assessment. The coefficient g is itself derived from the *elastic follow-up factor* r_M (RB 3643), a structural indicator of how much a locally

yielding cross-section is elastically restrained (“followed up”) by the surrounding, comparatively stiffer, structure.

This work has two objectives:

1. to present, in a self-contained manner, the historical and theoretical background that leads from classical beam theory to the modern treatment of elastic follow-up and the abatement coefficient in RCC-MRx;
2. to benchmark the RCC-MRx analytical procedure against a reference elasto-plastic finite-element (FE) shell analysis on a simple, well-defined elbow test case, in order to quantify how conservative (or non-conservative) the code procedure is with respect to a full non-linear reference solution.

2 Historical background

The treatment of piping flexibility and secondary stresses has evolved over more than two and a half centuries, from the classical beam theories of the 18th century to the elasto-plastic follow-up concepts introduced in the 1980s. Table 1 summarises the main milestones relevant to the present work, while Figure 1 visualises the same progression as it was synthesised in the source presentation.

Table 1: Historical milestones in beam theory and piping flexibility analysis.

Period	Contribution
1750	Euler–Bernoulli beam theory: classical elastic bending equations.
1911	Von Kármán’s work on the ovalisation of curved thin-walled tubes under bending [1]: first bend flexibility factors.
~1920	Timoshenko beam theory: shear deformation is included.
1940	Development of flexibility calculations for high-temperature steam piping (up to 650 °C) using graphical methods (Z-, L- and U-bend layouts).
1940s–1950s	A.R.C. Markl’s fatigue testing programme on piping components (Tube Turns) [2]: derivation of Stress Intensification Factors (SIFs) and generalisation of the flexibility-factor concept (building on von Kármán); this work forms the modern basis of piping flexibility and fatigue analysis.
~1980	R. Roche introduces the concept of elastic follow-up and of plasticity effects into piping flexibility calculations [4, 3].

The historical photographs complement the timeline by anchoring two strands of the story in concrete engineering practice: Figure 2 documents the experimental culture behind the modern SIF concept, while Figure 3 evokes the analytical framework from which the variational and energy-balance arguments used later in the paper emerged.

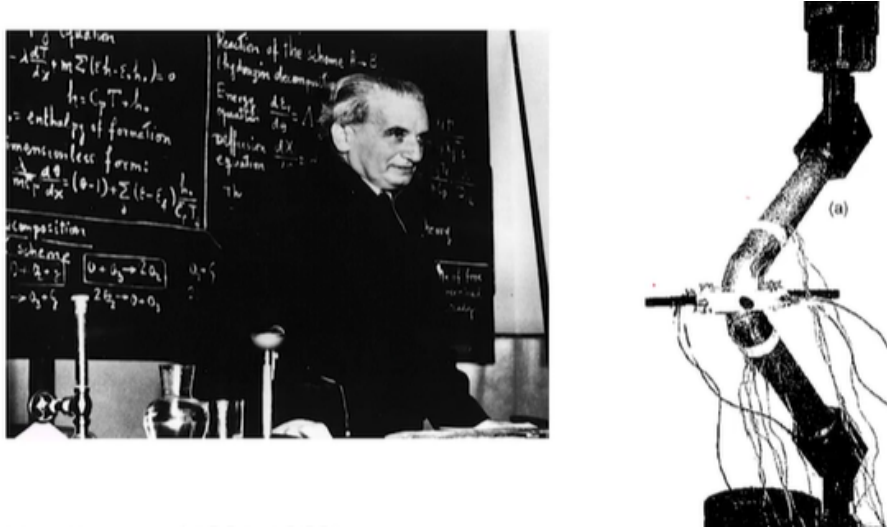


Figure 1: Timeline reproduced from the source presentation, linking the successive milestones of beam theory, elbow ovalisation, piping fatigue testing and elastic follow-up that culminate in the RCC-MRx treatment of displacement-controlled loads.

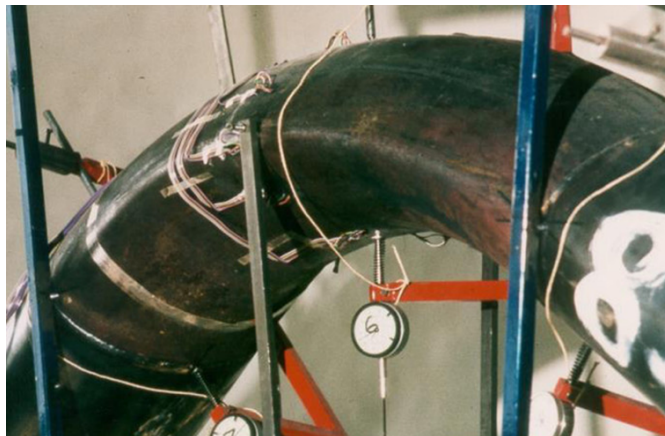


Figure 2: Photograph of one of A.R.C. Markl's original fatigue test rigs (Tube Turns, late 1940s–1950s), showing a pipe elbow specimen instrumented with dial gauges under cyclic bending [2].

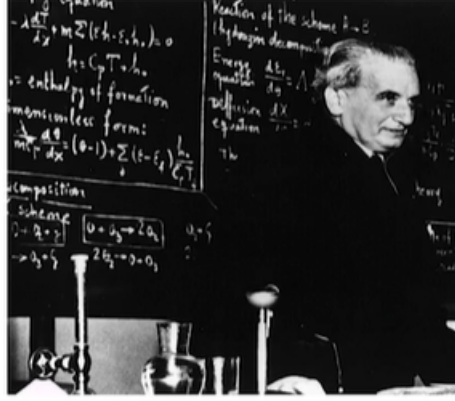


Figure 3: A lecturer at the blackboard, illustrating the enthalpy/reaction-scheme formalism typical of the mid-20th-century engineering-science teaching from which the beam-theory and thermodynamic energy-balance concepts used throughout this paper are inherited.

3 Beam theory and flexibility of curved pipes

3.1 Euler–Bernoulli and Timoshenko beam models

The classical Euler–Bernoulli hypothesis neglects transverse shear deformation and rotational inertia; it is only valid for slender members. The Timoshenko beam theory retains both transverse shear and all inertia terms and remains valid for stockier members as well. For in-plane bending of a beam in the (x, z) plane with constant cross-section and material along the longitudinal axis, the governing equations differ by the inclusion of (i) the rotational inertia associated with bending, (ii) the transverse shear flexibility (Timoshenko hypothesis), and (iii) the coupling between rotational and translational inertia [6]. Standard nuclear piping codes rely on Timoshenko-type beam elements for the straight and curved segments of the model. The corresponding governing partial differential equation for the transverse displacement $w(x, t)$ of a Timoshenko beam, including rotational inertia and shear-rotation inertia coupling, is given by

$$EI_y \frac{\partial^4 w}{\partial x^4} + \rho S \frac{\partial^2 w}{\partial t^2} - \rho I_y \left[1 + \frac{E}{k_z G} \right] \frac{\partial^4 w}{\partial x^2 \partial t^2} + \frac{\rho^2 I_y}{k_z G} \frac{\partial^4 w}{\partial t^4} - t_{z_{ext}} = 0, \quad (1)$$

where E is Young’s modulus, I_y the second moment of area, ρ the mass density, S the cross-sectional area, k_z the shear correction factor, G the shear modulus, and $t_{z_{ext}}$ the externally applied transverse load per unit length. When rotational inertia and shear-rotation inertia coupling are neglected (the classical Euler–Bernoulli specialisation, retaining only the distributed transverse load and the translational inertia term), Eq. (1) reduces to the familiar quasi-static beam equation used for the piping flexibility calculations of the remainder of this section,

$$\frac{\partial^2}{\partial x^2} \left(EI_y \frac{\partial^2 w}{\partial x^2} \right) + \rho S \frac{\partial^2 w}{\partial t^2} - t_{z_{ext}} = 0. \quad (2)$$

3.2 Elbow ovalisation and flexibility factors

Under in-plane bending, a curved pipe (elbow) undergoes cross-sectional *ovalisation*: the circular cross-section distorts into an oval shape. This ovalisation, illustrated by the shell result in Figure 5, increases the flexibility of the elbow well beyond the prediction of classical (straight) beam theory. The *flexibility factor* k is defined as the ratio between the rotation per unit length of the curved

component and the rotation per unit length of an equivalent straight pipe under the same bending moment, as illustrated in Figure 4: a straight length l under moment M rotates by θ , whereas the same length of curved component under the same moment rotates by $k\theta$.

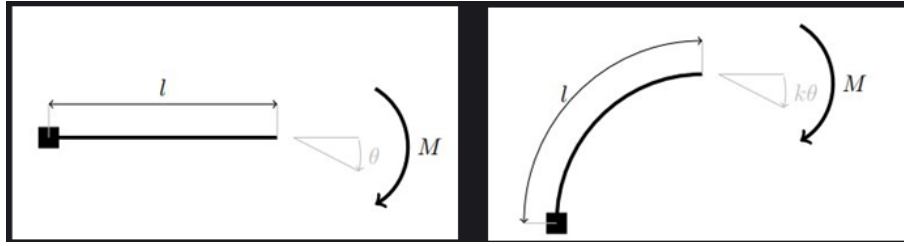


Figure 4: Graphical definition of the flexibility factor k , reproduced from the source presentation: a straight length l under end moment M rotates by θ (left), while the same length of curved component under the same moment rotates by $k\theta$ (right).

The source presentation first recalls the older von Kármán/Hovgaard elbow expression

$$k = \frac{12h^2 + 10}{12h^2 + 1}, \quad (3)$$

then introduces Beskin’s simpler reference approximation [1]. In the slide sequence, the latter is presented as the practical formula retained for piping calculations, without an explicit intermediate derivation from the former. The flexibility factor of a smooth, unpressurised elbow is therefore written in the benchmark, in terms of the non-dimensional flexibility characteristic h , as

$$h = \frac{tR}{r^2}, \quad k = \frac{1.65}{h} = \frac{1.65 r^2}{tR}, \quad (4)$$

where t is the wall thickness, r the mean cross-sectional radius, and R the bend radius.

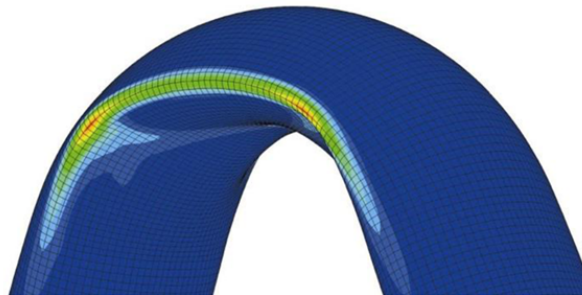


Figure 5: Finite-element illustration of elbow cross-section ovalisation under in-plane bending: the stress concentration follows the ovalised region at the crown of the bend.

For the elbow considered in the benchmark of this paper (outer diameter $D = 219.07$ mm, thickness $t = 3.76$ mm, bend radius $R = 500$ mm, hence $r = \frac{1}{2}(D - t) = 107.66$ mm), Eq. (4) gives $h = 0.1622$ and $k = 10.17$, in agreement with the value used in the benchmark (Section 7).

Link with Stress Intensification Factors (SIFs). Markl observed that the flexibility factor k and the stress intensification factor i (Section 4) are governed by the same underlying physics — the ovalisation of the cross-section — and that the inaccuracies affecting k and i tend to compensate one another. This observation justifies the continued use of simple, closed-form expressions such as Eq. (4) in modern piping codes.

4 Stress Intensification Factors and the Markl fatigue design curve

Because of the geometric and material discontinuities near fittings, welds and bends, the local stress in these areas is amplified with respect to the nominal beam-theory stress. This amplification is expressed through the *Stress Intensification Factor* (SIF), i . Following Markl's original definition, i is not simply a stress concentration factor but a fatigue *severity* factor: it quantifies how much the fatigue endurance of a given detail is degraded relative to a plain, defect-free straight pipe subjected to the same nominal bending stress.

Markl established an empirical fatigue (S – N) design curve from cyclic bending tests on straight, girth-welded pipe sections, of the form

$$S \cdot N^n = m, \quad (5)$$

where $S = iM/Z$ is the nominal bending stress amplitude multiplied by the SIF, Z the section modulus, N the number of cycles to failure, and m, n empirical constants fitted from the test data. Equation (5) is the compact algebraic counterpart of the fatigue-testing programme exemplified in Figure 2 (for austenitic stainless steels, $m \approx 1940$ MPa and $n \approx 0.2$).

Markl's equation

- ❑ The expansion moment range, as the pipe expands from cold (ambient) to hot (operating), must be limited such that

$$i \frac{M_E}{Z} \leq \left(f = \frac{6}{N^{0.2}} \leq 1.0 \right) (1.25 S_c + 0.25 S_h)$$

i = stress intensification factor; M_E = thermal expansion moment range (in.ln); Z = pipe section modulus (in³); N = number of cycles; S_c = allowable stress in the cold condition (psi); S_h = allowable stress in the hot condition (psi).

- ❖ $6/N^{0.2}$ is from Markl's relationship $iSN^{0.2} = C$.
- ❖ The limit on f is so that at less than 7,000 cycles the stress no longer is allowed to increase.
- ❖ $1.25S_c + 1.25S_h$ limit is to prevent ratcheting.
- ❖ $0.25S_h = 1.25S_h - 1S_h$ because Markl assumes that the pressure + deadweight stress is at its limit of S_h and deducts it.

Figure 6: Final technical frame of the historical video embedded in the source presentation: Markl's operating equation is written as $iM_E/Z \leq (f = 6/N^{0.2} \leq 1.0)(1.25S_c + 0.25S_h)$, i.e. the thermal-expansion moment range is limited by the fatigue-cycle factor f with an upper cap at 1.0. The accompanying bullets explain that the cap prevents the allowable stress range from increasing any further below about 7000 cycles, and that the $1.25S_c + 0.25S_h$ form is intended to prevent ratcheting after deducting the pressure-plus-deadweight share already assumed to consume the hot allowable.

A key methodological point made by Markl is the distinction between two possible references for the SIF:

- a *theoretical* reference, based on tests on ideal, notch-free homogeneous material, which yields comparatively high SIFs;
- the *as-built product* reference, based on tests on actual commercial, girth-welded pipe (including realistic surface imperfections), which is the reference actually adopted by the design rules.

This choice of reference explains why SIFs measured in fatigue tests are systematically lower than those predicted by purely elastic (theoretical) stress-concentration analysis, and it underlies the overall internal consistency of Markl’s approach.

In practice, this matters here because the largest SIFs are usually found precisely in the neighbourhood of geometrical discontinuities (elbows, branch connections, fittings, weld transitions), i.e. where local plastic strains are also most likely to develop under imposed displacements. The source presentation explicitly uses this observation as the bridge between the classical fatigue/SIF framework and the RCC-MRx abatement framework: the same locations that govern fatigue severity are natural candidates for elastic follow-up and secondary-stress abatement. It also notes that, at high temperature and over long durations, creep and stress relaxation can further modify this redistribution; those time-dependent effects are outside the scope of the present benchmark, which is restricted to monotonic elastic-plastic follow-up.

5 Elastic follow-up: from local (Neuber) to global (Roche) redistribution

Two conceptually different, but often confused, corrections are used to relate an elastically computed stress or strain field to the actual elasto-plastic response of a structure: *Neuber’s rule* and the *elastic follow-up effect*. Their essential differences are summarised in Table 2. The source presentation briefly mentions both r_M and r_C as follow-up-related indicators; only the RCC-MRx factor r_M is developed here because it is the quantity actually used in clauses RB 3643 and RB 3651 and in the benchmark calculations.

Table 2: Neuber local correction versus global elastic follow-up.

	Neuber	Elastic follow-up
Nature	Local	Global
Affected zone	Notch / point	Whole structure
Type of phenomenon	Stress concentration	Strain redistribution
Role	Corrects a local stress peak	Corrects the overall structural response

Elastic follow-up describes how a locally yielding (softening) section of a structure is “followed up” by the surrounding, stiffer parts, which continue to impose additional deformation on the yielded section because the imposed *overall* displacement must still be accommodated. Depending on how stiff the surrounding structure is relative to the yielding section, the behaviour of that section under a nominally secondary (displacement-controlled) load can range from purely secondary (self-limiting, if the surrounding structure is very flexible and simply relaxes) to effectively primary (if the surrounding structure is very stiff and keeps imposing the full displacement onto the yielded, and therefore weaker, section). Figure 7 gives the constitutive interpretation of these two limiting paths.

Slides 16–17 of the source presentation illustrate this contrast quantitatively on a small didactic DN200 bar example, explicitly built so that a force-controlled case (Bar 1) and a displacement-controlled case (Bar 2) share the same elastic starting point. The common geometry is a DN200 tube of length 2 m and thickness 3.76 mm, with $E = 155$ GPa and $\nu = 0.3$. Bar 1 is loaded by $F_1 = 320$ kN, giving $\sigma_{el} = F_1/S = 126$ MPa and $\varepsilon_{el} = \sigma_{el}/E = 0.081$ %. Bar 2 is then assigned an imposed displacement $u_2 = 1.623$ mm so that the same elastic stress is recovered independently through $\sigma_{el} = (u_2/L)E = 126$ MPa; Figure 8 reproduces the slide’s graphical construction (slide 16) of this common elastic point $(\sigma_{el}, \varepsilon_{el})$ on the Hookean line $E = \sigma/\varepsilon$, shared by both bars before plasticity is introduced.

Once plasticity is admitted (slide 17), the primary path reaches $\sigma_1 = 126$ MPa and $\varepsilon_1 = 1$ %, whereas the purely secondary path relaxes to $\sigma_2 = 82.3$ MPa and $\varepsilon_2 = 0.081$ % – the exact

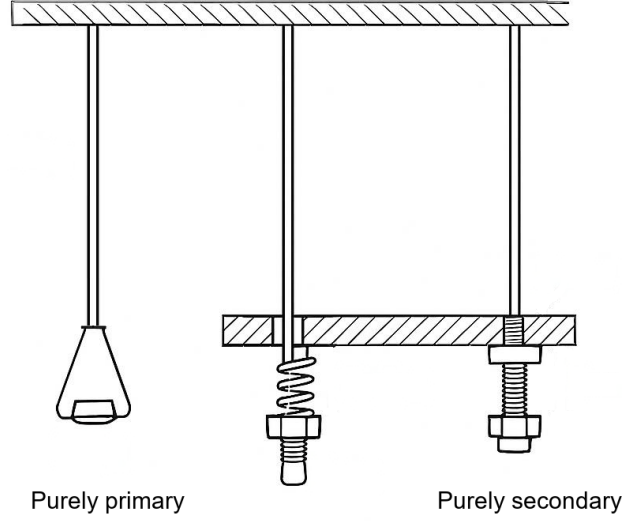


Figure 7: Source-presentation construction contrasting the Ramberg–Osgood response of a purely primary loading path with that of a purely secondary loading path: the former keeps accumulating strain at nearly unchanged stress, whereas the latter relaxes stress once plasticity develops.

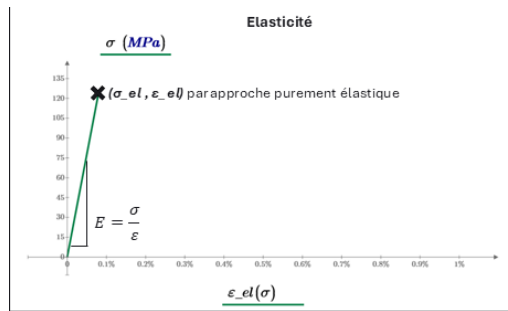


Figure 8: Slide-level graphical construction (slide 16) of the common elastic reference point $(\sigma_{el}, \varepsilon_{el}) = (126 \text{ MPa}, 0.081 \%)$ shared by Bars 1 and 2, obtained independently from F_1/S (Bar 1) and $(u_2/L)E$ (Bar 2) before the elasto-plastic calculations of Figure 10 are run.

pedagogical contrast used on the slides. Figure 9 plots Bars 1 and 2’s elastic and elasto-plastic points $(\sigma_{el}, \varepsilon_{el})$, $(\sigma_1, \varepsilon_1)$ and $(\sigma_2, \varepsilon_2)$ directly on the same Ramberg–Osgood curve used for the DN200 example, before Bar 3 (the spring case discussed in Section 6) is introduced.

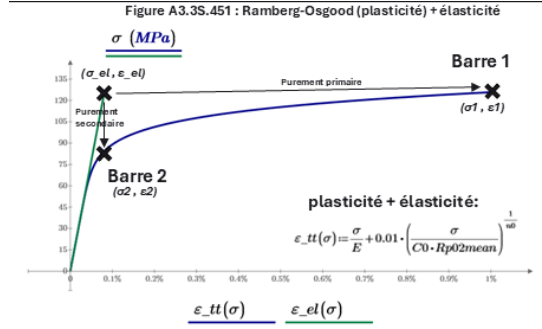


Figure 9: Ramberg–Osgood curve (RCC-MRx Figure A3.3S.451) with Bar 1’s purely primary point $(\sigma_1, \varepsilon_1)$ and Bar 2’s purely secondary point $(\sigma_2, \varepsilon_2)$ located relative to the common elastic point $(\sigma_{el}, \varepsilon_{el})$, reproduced from the presentation.

Table 3: Worked DN200 example reproduced from the presentation to contrast primary and secondary loading paths.

Quantity	Value	Role in the example
Nominal size / geometry	DN200, $L = 2$ m, $t = 3.76$ mm	common bar geometry
Elastic constants	$E = 155$ GPa, $\nu = 0.3$	RCC-MRx A3.3S.451 data used on the slide
Primary input	$F_1 = 320$ kN	force applied to Bar 1
Secondary input	$u_2 = 1.623$ mm	imposed end displacement applied to Bar 2
Common elastic stress	$\sigma_{el} = 126$ MPa	obtained both from F_1/S and from $(u_2/L)E$
Common elastic strain	$\varepsilon_{el} = 0.081$ %	elastic starting point of both paths
Primary elasto-plastic point	$(\sigma_1, \varepsilon_1) = (126 \text{ MPa}, 1 \text{ %})$	force-controlled response
Secondary elasto-plastic point	$(\sigma_2, \varepsilon_2) = (82.3 \text{ MPa}, 0.081 \text{ %})$	self-limiting displacement-controlled response

The corresponding FE stress–strain calculations (linear-elastic and elasto-plastic MSC Marc runs on the two bars) are reproduced in Figure 10: because Bars 1 and 2 are calibrated to the same elastic stress, their elastic contour is identical, while the elasto-plastic contours diverge exactly as summarised in Table 3 – Bar 1 remains at the primary stress level (126.34 MPa), whereas Bar 2 relaxes to the secondary level (82.3 MPa).

5.1 The Ramberg–Osgood constitutive law

To model the elasto-plastic material response, the Ramberg–Osgood law is used in its general textbook form, decomposing the total strain into an elastic and a power-law plastic part,

$$\varepsilon_t = \varepsilon_e + \varepsilon_p = \frac{\sigma}{E} + \left(\frac{\sigma}{K} \right)^{1/n}, \quad (6)$$

with K a strength coefficient and n a hardening exponent. The source presentation specialises Eq. (6) to RCC-MRx’s own parametrisation of the strength coefficient as a temperature-dependent multiple of the 0.2% proof stress,

$$K(T) = C_0 R_{p0.2}(T), \quad (7)$$

and to the code’s own definition of the plastic-strain scale (the leading 0.01, or 100×, factor below reflects the code’s percent-strain convention), giving the exact form used in the benchmark

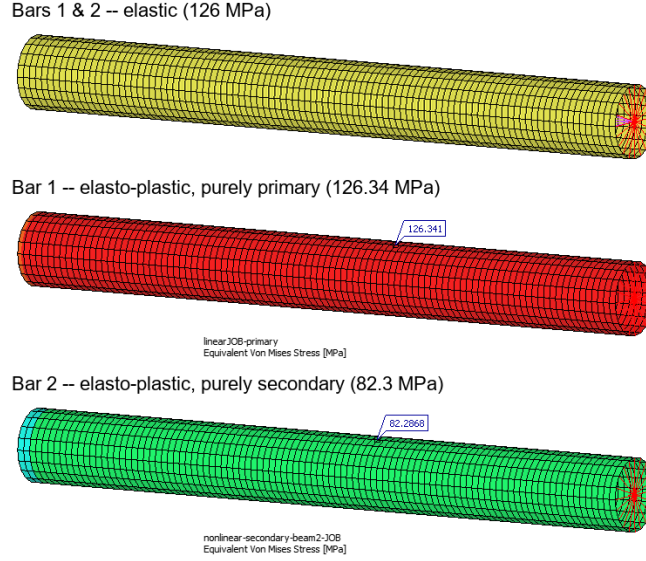


Figure 10: MSC Marc equivalent von Mises stress contours for the DN200 toy example. Top: common elastic state shared by Bars 1 and 2 (126 MPa). Middle: Bar 1’s elasto-plastic, purely primary response (*linearJOB-primary*), remaining close to the elastic stress level. Bottom: Bar 2’s elasto-plastic, purely secondary response (*nonlinear-secondary-beam2-JOB*), relaxing to a markedly lower stress – the self-limiting behaviour characteristic of displacement-controlled loading.

computations:

$$\varepsilon(\sigma) = \frac{\sigma}{E} + 0.01 \left(\frac{\sigma}{C_0 R_{p0.2, \text{mean}}} \right)^{1/n_0}, \quad (8)$$

with, for 316L(N) stainless steel, $n_0 = 0.1125$ ($1/n_0 = 8.89$) and $C_0 = 1.198$; at the benchmark temperature of 550°C , $R_{p0.2} \approx 106$ MPa, so that the stress scale $C_0 R_{p0.2}$ is about 131.8 MPa. The temperature dependence therefore enters through $R_{p0.2}(T)$ in Eq. (7), whereas C_0 and n_0 retain the same normalised curve shape at all temperatures — i.e., no change of hardening *shape* with temperature is assumed. A sensitivity study reported in the presentation, using an alternative parameter set ($C_0 = 1.233$, $n_0 = 0.130$, i.e. $1/n_0 = 7.69$, $C_0 R_{p0.2} \approx 135.6$ MPa) calibrated on the same 550°C tensile data, produces a stress–strain curve only marginally softer than the reference one (e.g. about 2 MPa lower at $\varepsilon_t = 2\%$), confirming that the choice of hardening parameters is not the dominant source of discrepancy discussed in Section 10.

Several open questions regarding the domain of validity of this simplified, temperature-invariant hardening shape are noted in the presentation (and are only partially addressed here): the sensitivity to strain rate, to monotonic versus cyclic hardening, to the number of cycles, and to environment (air, PWR primary water, sodium, etc.). Figure 11 documents both the temperature family and the parameter sensitivity used to motivate this discussion. For reference, an example comparison at 0.01 true strain shows a stress discrepancy of about 4.0 MPa ($\sim 5\%$) between candidate hardening curves [7].

5.2 Roche’s spring-bar analogy

R. Roche’s original demonstration of elastic follow-up [4] uses a simple three-bar analogy:

- **Bar 1** is subjected to a purely primary load: the elastically computed stress–strain state $(\sigma_{\text{el}}, \varepsilon_{\text{el}})$ moves, once plasticity is included, to $(\sigma_1, \varepsilon_1)$ on the Ramberg–Osgood curve, following the classical, dangerous, primary-load behaviour (unbounded deformation for stresses above yield).

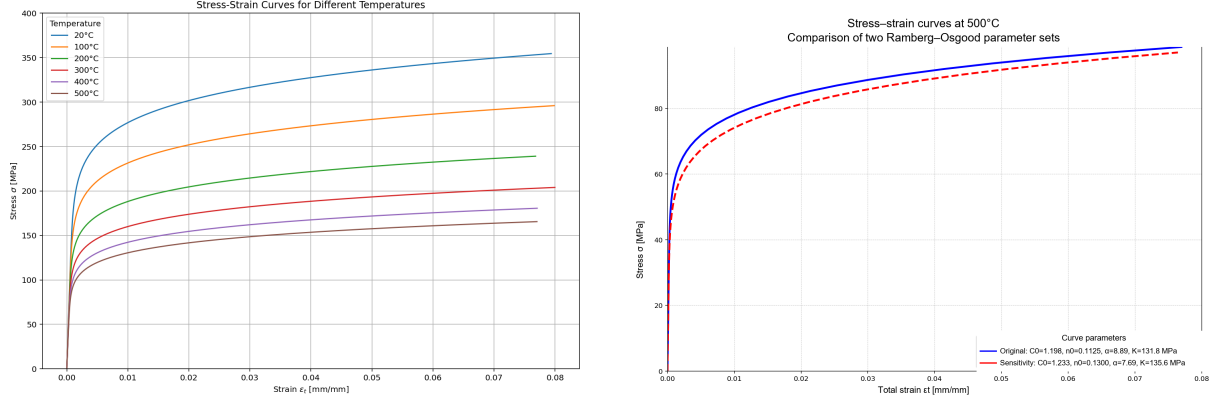


Figure 11: Ramberg–Osgood material data extracted from the presentation. Left: family of 316L(N) stress–strain curves used to justify the temperature-dependent scaling of the law through $R_{p0.2}(T)$. Right: sensitivity check at 500°C–550°C, comparing the reference parameter set ($C_0 = 1.198$, $n_0 = 0.1125$) with the alternative fit ($C_0 = 1.233$, $n_0 = 0.130$) discussed in the text.

- **Bar 2** is subjected to a purely secondary (displacement-controlled) load: with plasticity, the elastic state $(\sigma_{el}, \epsilon_{el})$ moves to (σ_2, ϵ_2) , exhibiting the self-limiting character typical of secondary stresses.
- **Bar 3** reproduces Bar 2’s geometry and loading but adds a *spring* in series, with an imposed end displacement calibrated so that the resulting *elastic* stress equals that of Bars 1 and 2. Despite representing a nominally purely secondary load case, Bar 3 exhibits an intermediate behaviour, between that of Bar 1 (primary) and Bar 2 (secondary): (σ_3, ϵ_3) lies between (σ_1, ϵ_1) and (σ_2, ϵ_2) .

This example shows that a nominally secondary stress can, in the presence of elastic follow-up (represented here by the series spring), acquire a partly primary character. A parametric study of the spring stiffness k_2 confirms this trend continuously: as $k_2 \rightarrow \infty$, Bar 3 tends towards Bar 2’s purely secondary behaviour; as $k_2 \rightarrow 0$, Bar 3 tends towards Bar 1’s primary-like behaviour. Figure 12 shows the conceptual three-bar construction, Figure 14 its FE counterpart, Figure 16 the spring-stiffness sweep extracted from the presentation, and Figure 17 an original numerical example from Roche. RCC-MRx later formalised this behaviour through the elastic follow-up factor r_M and the abatement coefficient g (Section 6).

A three-bar FE validation of this analogy, carried out with MSC Marc on bars of unit cross-section, gives an elastic reference stress $\sigma_{el} = 126$ MPa; the elasto-plastic calculations give $\sigma_1 = 126.34$ MPa for the purely primary Bar 1 (*linear-primary* job) and $\sigma_2 = 82.3$ MPa for the purely secondary Bar 2 (*nonlinear-secondary-beam2* job). Figure 13 first places Bar 3’s single representative point (σ_3, ϵ_3) on the same curve, visibly intermediate between Bar 1 and Bar 2, before the full parametric sweep is examined. For Bar 3, a parametric study on the series-spring stiffness k_2 confirms the continuous transition announced above: the extracted points A–D of Figure 16 correspond to $k_2 = 2k, k/2, k/4$ and $k/16$, for which r_M rises from 0.996 to 3.989, 7.981 and 31.931, while the estimated stress increases from $\sigma_{3,est} = 89.912$ MPa, 97.183 MPa, 101.8 MPa and 111.4 MPa and the associated strain from $\epsilon_{3,est} = 0.104\%$, 0.155%, 0.206% and 0.383%. The same sweep shows the end displacement increasing from 2.435 mm to 4.869 mm, 8.115 mm and 27.591 mm, i.e. Bar 3 moves away from the purely secondary limit and towards the primary-like response as the spring restraint is relaxed.

The presentation also documents Bar 3’s FE contours at another point of the spring-stiffness sweep, reproduced in Figure 15: the elastic reference state (126.056 MPa, essentially identical to Bars 1 and 2 since the spring is calibrated to preserve the elastic stress) and a further elasto-plastic run at 95.852 MPa, again located between Bar 1’s and Bar 2’s limiting responses – consistent

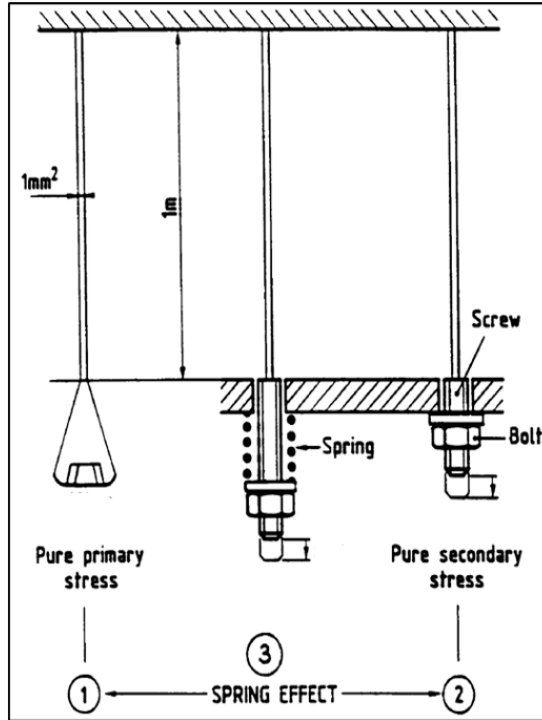


Figure 12: Roche's original spring-bar analogy: Bar 1 (left) carries a pure primary stress (dead-weight-like load); Bar 2 (centre) carries a pure secondary stress through a soft spring; Bar 3 (right) reproduces the secondary loading through a bolted, effectively rigid connection, producing an intermediate, “spring effect” behaviour between Bars 1 and 2.

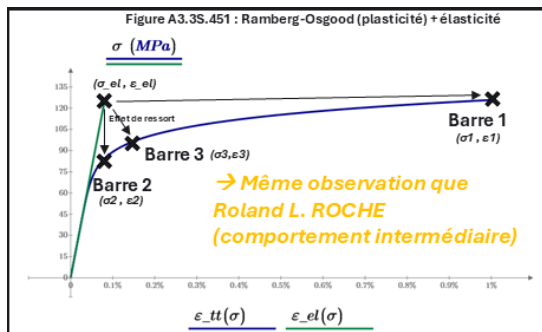


Figure 13: Ramberg–Osgood curve with Bar 3's point $(\sigma_3, \varepsilon_3)$ (“Effet de ressort”) added to Bars 1 and 2, showing at a glance the intermediate character of the spring-effect response, before the parametric sweep of Figure 16 is introduced.

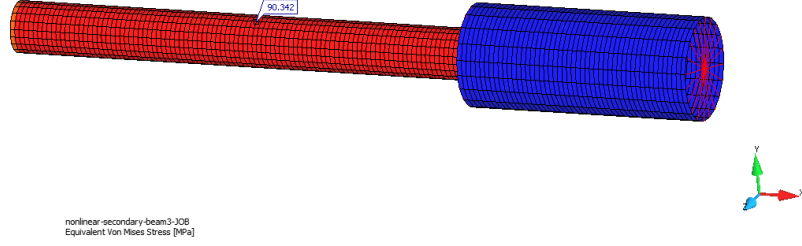


Figure 14: MSC Marc equivalent von Mises stress contour for Bar 3 (*nonlinear-secondary-beam3* job): the stress in the flexible segment reaches 90.342 MPa, an intermediate value between Bar 1’s primary stress ($\sigma_1 = 126.34$ MPa) and Bar 2’s secondary stress ($\sigma_2 = 82.3$ MPa), illustrating the “spring effect” for a representative spring stiffness k_2 .

with, though not numerically identical to, the closest tabulated estimate ($\sigma_{3,\text{est}} = 101.8$ MPa at $u_3 = 8.115$ mm in Figure 16), the two figures reflecting slightly different runs of the same parametric family.

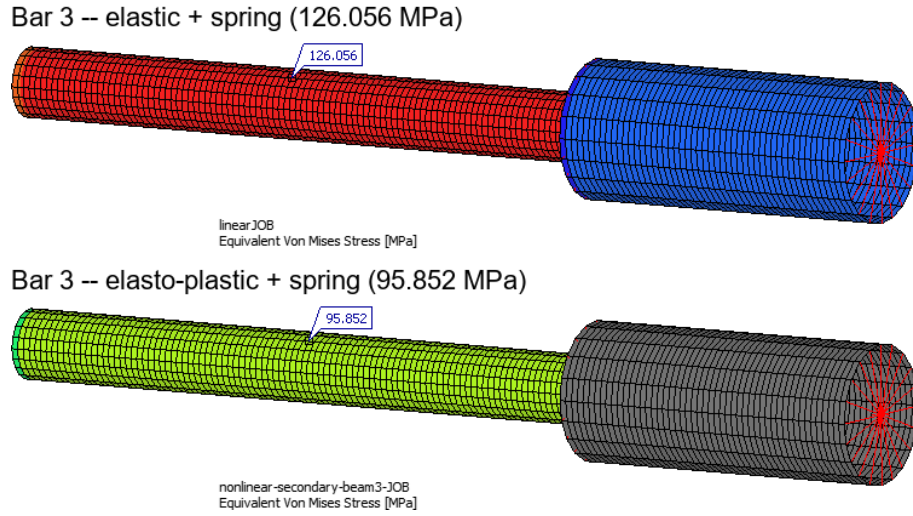


Figure 15: MSC Marc equivalent von Mises stress contours for a second, independent spring-stiffness run on Bar 3: elastic reference state (top, 126.056 MPa) and elasto-plastic response (bottom, 95.852 MPa), both showing the flexible bar segment together with the stiffer series-spring segment.

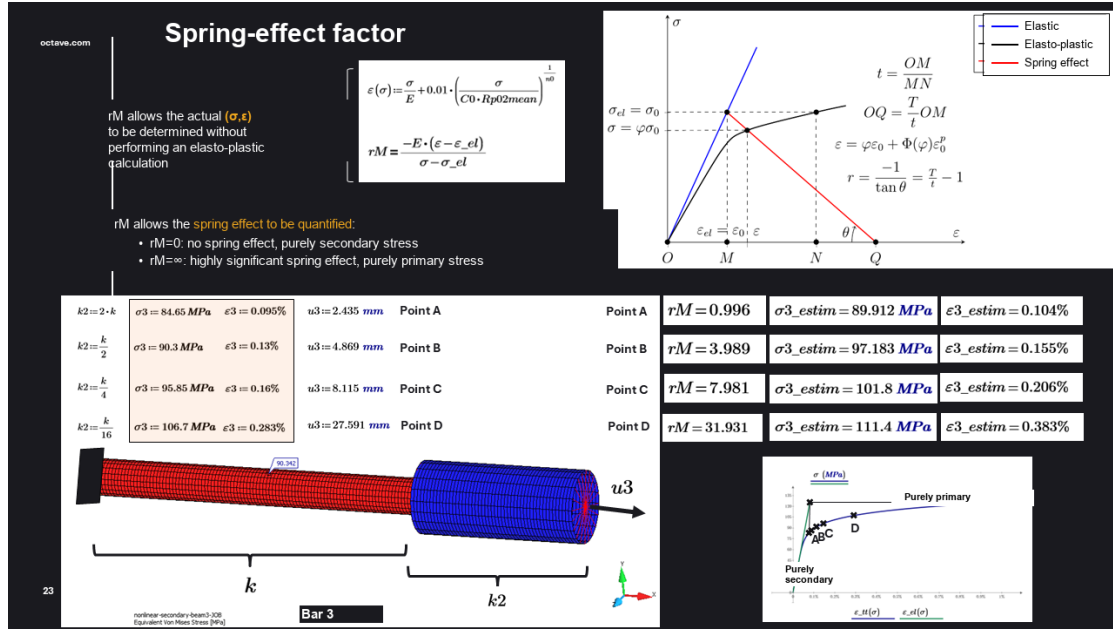


Figure 16: Parametric three-bar sweep extracted from the presentation: reducing the added spring stiffness from $k_2 = 2k$ to $k_2 = k/16$ drives Bar 3 continuously away from the purely secondary limit (Bar 2) and towards the primary-like limit (Bar 1). The slide reports the corresponding sequence of follow-up factors ($r_M = 0.996, 3.989, 7.981, 31.931$), stress estimates ($\sigma_{3,est} = 89.912 \text{ MPa}, 97.183 \text{ MPa}, 101.8 \text{ MPa}$ and 111.4 MPa) and end displacements ($u_3 = 2.435 \text{ mm}, 4.869 \text{ mm}, 8.115 \text{ mm}$ and 27.591 mm).

$L := 100 \text{ in} = 2.54 \text{ m}$
 $E := 192000 \text{ MPa}$

$L_{faible} := 32 \text{ in} = 0.813 \text{ m}$
 $K := 793 \text{ MPa}$

$L_{forte} := L - L_{faible} = 1.727 \text{ m}$
 $m := \frac{1}{2.915}$

$$\Delta \epsilon (\Delta \sigma) := \frac{\Delta \sigma}{E} + 0.01 \cdot \left(\frac{\Delta \sigma}{K} \right)^{\frac{1}{m}}$$

Figure 17: Numerical data from Roland Roche's original worked example (straight bar of total length $L = 2.54 \text{ m}$, split into a flexible segment $L_{weak} = 0.813 \text{ m}$ and a stiff segment $L_{stiff} = 1.727 \text{ m}$, with $E = 192\,000 \text{ MPa}$, Ramberg–Osgood parameters $K = 793 \text{ MPa}$, $m = 1/2.915$), illustrating the calculation of the elastic follow-up factor on a simple two-segment bar.

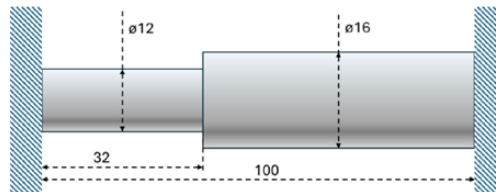


Figure 18: Geometry of Roland Roche's original two-segment bar example: a stepped shaft with a flexible segment of diameter 12 mm and length 32 in and a stiffer segment of diameter 16 mm, clamped at both ends, with the total length (100 in) measured from the flexible end.

6 The RCC-MRx elastic follow-up factor and abatement coefficient

6.1 Energy balance and definition of the elastic follow-up factor

Before stating the RCC-MRx result, it is useful to make explicit the two independent hypotheses combined by the presentation to reach it, since they are attributed to two different authors and are easily conflated. The material non-linearity is carried by the Ramberg–Osgood strain decomposition, written here in incremental form relative to a reference elastic state (subscript 0),

$$\Delta \bar{\varepsilon} = \varphi \Delta \bar{\varepsilon}_0 + \Phi(\varphi) \Delta \bar{\varepsilon}_0^P, \quad (9)$$

while the loading-path non-linearity is carried by Kachanov’s hypothesis of a radially proportional (self-similar) stress redistribution,

$$\Delta \bar{\sigma} = \varphi \Delta \bar{\sigma}_0, \quad (10)$$

with the same scalar multiplier φ in both relations, and $\Phi(\varphi) = \varphi^{1/n_0}$ the Ramberg–Osgood exponent function of Eq. (8). Equation (10) – i.e., that the elasto-plastic stress field remains everywhere collinear with the purely elastic one, only rescaled by φ – is Kachanov’s contribution and is the key simplifying assumption that turns a genuinely non-linear redistribution problem into a scalar one; Eq. (9) is the Ramberg–Osgood material response evaluated along that self-similar path. Writing the virtual-work statement of the elasto-plastic increment against the elastic reference field, and separating an imposed-displacement contribution Δu (secondary) from a thermal/imposed-strain contribution $\Delta \bar{\varepsilon}$,

$$\varphi \int \Delta \bar{\varepsilon}_0 : \Delta \bar{\sigma}_0 dv + \Phi(\varphi) \int \Delta \bar{\varepsilon}_0^P : \Delta \bar{\sigma}_0 dv = \int \Delta u \Delta F_0 ds + \int \Delta \bar{\varepsilon} : \Delta \bar{\sigma}_0 dv, \quad (11)$$

and comparing with the energy balance of the purely elastic calculation itself (same loading, $\varphi = \Phi = 1$),

$$\frac{1}{2} \int \Delta \bar{\varepsilon}_0 : \Delta \bar{\sigma}_0 dv = \frac{1}{2} \int \Delta u \Delta F_0 ds + \frac{1}{2} \int \Delta \bar{\varepsilon} : \Delta \bar{\sigma}_0 dv, \quad (12)$$

subtracting Eq. (12) from Eq. (11) eliminates the (generally unknown) right-hand-side loading terms and leaves a purely field-integral relation,

$$(\varphi - 1) \int \Delta \bar{\varepsilon}_0 : \Delta \bar{\sigma}_0 dv + \Phi(\varphi) \int \Delta \bar{\varepsilon}_0^P : \Delta \bar{\sigma}_0 dv = 0 \implies (\varphi - 1) T + \Phi(\varphi) = 0, \quad \frac{1}{T} = \frac{\int \Delta \bar{\varepsilon}_0^P : \Delta \bar{\sigma}_0 dv}{\int \Delta \bar{\varepsilon}_0 : \Delta \bar{\sigma}_0 dv}, \quad (13)$$

which defines the global resilience T purely from the elastic solution, without solving the non-linear problem. Passing to equivalent (von Mises) stress and strain increments $\Delta \varepsilon^*$, $\Delta \sigma^*$ as in Eq. (34), and using the elastic relation $\Delta \varepsilon_0^* = 2(1 + \nu) \Delta \sigma_0^* / 3E$, Eq. (13) reduces to the compact result already stated in Eq. (39) of Appendix B,

$$\frac{3}{2(1 + \nu)} \cdot \frac{E (\Delta \varepsilon^* - \Delta \varepsilon_0^*)}{\Delta \sigma_0^* - \Delta \sigma^*} = \frac{T}{t} - 1, \quad (14)$$

i.e. the same elastic follow-up factor r_M recovered below from Roche’s total-quantity (rather than incremental) notation; the full step-by-step derivation, including the definition of the local resilience t , is reproduced in Appendix B.

RCC-MRx clause RB 3643 formalises the elastic follow-up factor r_M from an energy balance of the purely elastic calculation, following R. Roche’s original derivation (reproduced in full in Appendix B). Let $(\sigma_{el}, \varepsilon_{el})$ be the elastically computed equivalent stress and strain at the section of interest, and (σ, ε) the actual (elasto-plastic) equivalent stress and strain reached at the same

section once the true, non-linear material law (here, Ramberg–Osgood, Eq. (8)) is accounted for, with the loading history assumed radially proportional ($\bar{\sigma} = \varphi \bar{\sigma}_0$, i.e. the elasto-plastic stress tensor field remains everywhere proportional to the elastic one through a single scalar φ). RCC-MRx defines the elastic follow-up factor as

$$r_M = -\frac{E(\varepsilon - \varepsilon_{\text{el}})}{\sigma - \sigma_{\text{el}}} = \frac{T}{t} - 1 \geq 0, \quad (15)$$

where t is the *local* resilience (degree of elasticity) of the section under consideration, and T is a *global* resilience computed over the whole structure by weighting every section by its elastic reference stress:

$$T = \frac{\int \sigma_{\text{ref}}^2 A ds}{\int \frac{\sigma_{\text{ref}}^2 A}{t} ds}, \quad \frac{1}{t} = \frac{\varepsilon_{M,p}(\sigma_{\text{ref}})}{\sigma_{\text{ref}}/E}, \quad \sigma_{\text{ref}}^2 = \left[0.79 B_2 \frac{(m)'_R}{Z} \right]^2 + \left[0.87 \frac{m_1}{Z} \right]^2, \quad (16)$$

with $\varepsilon_{M,p}(\sigma) = 0.01 [\sigma / (C_0 R_{p0.2, \text{mean}})]^{1/n_0}$ the plastic part of the mean Ramberg–Osgood curve (Eq. (8)), and m_1 , $(m)'_R$ the primary and secondary (resultant) moments at the section, Z its section modulus. Equation (16) is the practical, piping-code form of T ; its rigorous derivation from the energy balance $\iiint \bar{\sigma}_0 : \bar{\varepsilon} dV = \iiint \bar{\sigma}_0 : \bar{\varepsilon}_0 dV$ is given in Appendix B, Eqs. (13)–(14) and (22).

For isotropic materials, the rigorous derivation of Appendix B gives $r_M = \frac{2(1+\nu)}{3} (\frac{T}{t} - 1)$, which for steels ($\nu = 0.3$) gives $r_M = 0.867 (\frac{T}{t} - 1)$. In a conservative manner, RCC-MRx takes $\frac{2(1+\nu)}{3} \approx 1$ (absorbing the 3-D correction into a separate stress index of 0.87 applied to pressure and torsional moments), so that the code ultimately retains the simple form of Eq. (15), $r_M = T/t - 1$.

The factor r_M has the following limiting behaviour:

- $r_M = 0$ ($T = t$): no elastic follow-up; the stress is purely secondary (fully self-limiting).
- $r_M \rightarrow \infty$ ($T \gg t$): very significant elastic follow-up; the stress behaves as purely primary.

Note that a *higher* r_M is treated by the code as a *more conservative*, more primary-like, assumption; taking $r_M(\text{coeff} = 1) > r_M(\text{coeff} = 0.867)$ is therefore itself a conservative simplification.

In the Roche three-bar analogy, Bar 2 (no spring) corresponds to $r_M = 0$, Bar 1 (rigid connection) corresponds to $r_M \rightarrow \infty$, and Bar 3 (elastic spring) lies at an intermediate, finite value of r_M that depends on the spring stiffness k_2 .

Operationally, the presentation summarises the RCC-MRx workflow in four steps: (i) perform a purely elastic piping calculation; (ii) deduce the local plastic strain function $\varepsilon_{M,p}(\sigma_{\text{ref}})$ from the adopted Ramberg–Osgood law; (iii) compute the global resilience T by stress-squared weighting of the elastic reference field over the whole structure; and (iv) retain the governing local value of r_M (and therefore of g) at the critical section. Figures 19 and 20 summarise these complementary viewpoints: a graphical construction on the constitutive curve and an energy-balance route derived from the elastic FE solution. In other words, the code is intended to recover an elasto-plastic *estimate* from an elastic base calculation rather than to replace full non-linear analysis in all circumstances.

6.2 Definition of the abatement coefficient

For a section without internal pressure, RCC-MRx clause RB 3651.1131 defines the abatement coefficient as the ratio $g = \sigma / \sigma_{\text{el}}$ of the true elasto-plastic stress to the elastically computed stress at the same imposed strain. Figure 21 reproduces the code’s graphical construction, and the same result can be written directly from r_M and the Ramberg–Osgood law as

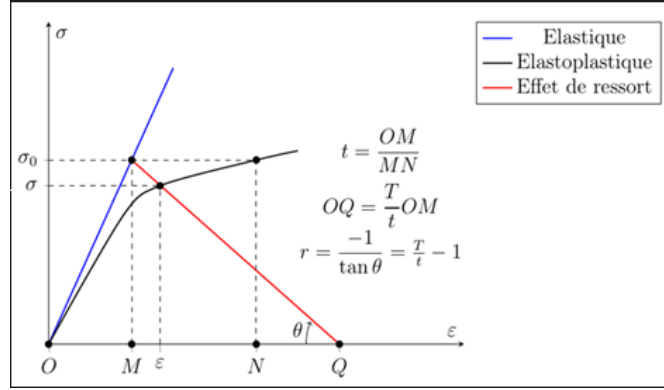


Figure 19: Graphical construction of r_M , reproduced directly from the presentation (a redrawn version was found to mislabel the relative positions of σ_0 and σ and has been replaced by this faithful crop): $\sigma_0 = \sigma_{el}$ is the elastic reference stress at point M , σ the actual elasto-plastic stress reached at the same strain ε ; the local resilience is $t = OM/MN$, the global resilience T is such that $OQ = (T/t) OM$, and the spring-effect line of slope $-1/r_M = \tan \theta$ passes through M and Q .

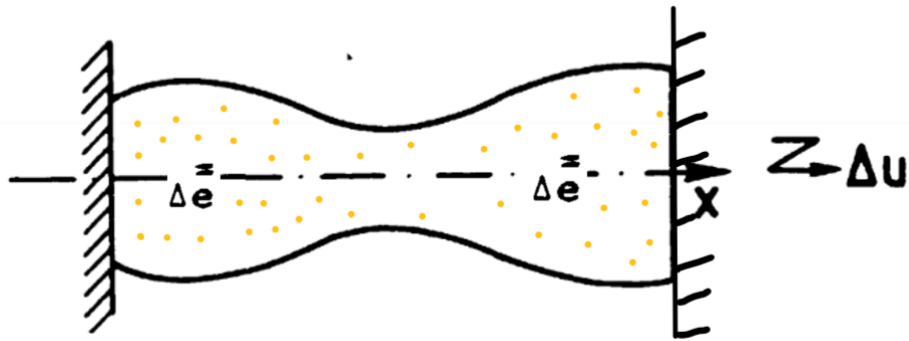


Figure 20: Energy-balance scheme extracted from the presentation, summarising Roche's route from the virtual-work statement of the elastic calculation to the equivalent stress/strain formulation used to define T , t and ultimately r_M .

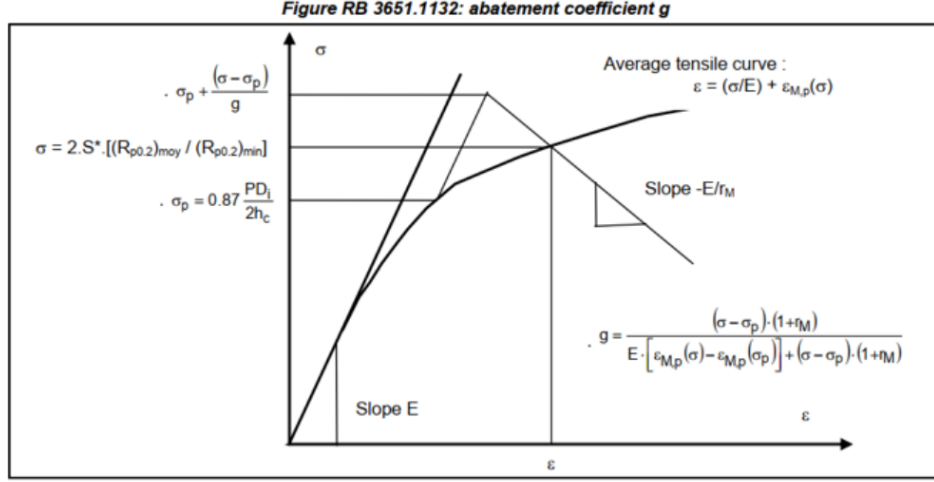


Figure 21: RCC-MRx Figure RB 3651.1132 (reproduced from the code): graphical definition of the abatement coefficient g , showing the average Ramberg–Osgood tensile curve, the elastic (slope E) and elastic-follow-up (slope $-E/r_M$) construction lines, the primary membrane stress σ_p and allowable stress S^* .

$$g = \frac{\sigma (1 + r_M)}{E \varepsilon_{M,p}(\sigma) + \sigma (1 + r_M)}. \quad (17)$$

When a primary membrane stress from pressure, $\sigma_P = 0.87 \frac{PD_i}{2h_c}$, is also present at the section (with $\sigma_P < \sigma$), the code generalises Eq. (17) to

$$g = \frac{(\sigma - \sigma_P)(1 + r_M)}{E [\varepsilon_{M,p}(\sigma) - \varepsilon_{M,p}(\sigma_P)] + (\sigma - \sigma_P)(1 + r_M)}. \quad (18)$$

The coefficient g is then used to reduce (“abate”) the displacement-dependent moment m (secondary, in the classical sense) into an abated value $g m$, which is combined with the primary loads in the assessment of primary-type (P-type) damage through the RCC-MRx CMS1 stress-limitation rule (RB 3651.1131):

$$\overline{s_m + s_b} = \left[\left(D_1 \left[\frac{PD_i}{2h_c} \right] \right)^2 + \langle D_2/Z, M + g \cdot m \rangle^2 \right]^{0.5} \leq S^*. \quad (19)$$

In this way, the combined effects of elastic follow-up, plasticity and elasticity are accounted for in the P-type damage check of RCC-MRx, via $\overline{s_m + s_b}$, without requiring the analyst to perform a full elasto-plastic finite-element calculation. The presentation motivates exactly this rule by asking whether secondary loads must be considered when validating primary stresses: the answer is illustrated by re-using the Bar 1/Bar 2/Bar 3 Ramberg–Osgood curve already given in Figure 16 (bottom-right panel) – since Bar 3’s response (points A–D) sits strictly between the purely primary (Bar 1) and purely secondary (Bar 2) limits, a nominally secondary moment can behave partly as a primary one, which is precisely why $g \cdot m$, rather than m alone, is added inside the CMS1 check of Eq. (19).

For the benchmark section reported later in Figure 34, the code output gives $\sigma = 119.59$ MPa, $\varepsilon_{M,p}(\sigma) = 0.587\%$ and $r_M = 2.638$. Substituting these values into Eq. (17) with $E = 155\,000$ MPa yields

$$g = \frac{119.59(1 + 2.638)}{155000 \times 0.00587 + 119.59(1 + 2.638)} = 0.3237,$$

which gives $g \approx 0.3235$, i.e. the reported production-code value $g = 0.3237$ to within rounding of the tabulated strain. This direct numerical check was used here as an internal verification of the transcribed RCC-MRx formulae.

Table 4: Consolidated notation for the RCC-MRx elastic-follow-up route.

Quantity	Definition / role
t	Local resilience of the section, defined through $1/t = \varepsilon_{M,p}(\sigma_{\text{ref}})/(\sigma_{\text{ref}}/E)$; low t means a locally more plasticised section.
T	Global resilience of the whole structure, obtained by stress-squared weighting of all sections through Eq. (16).
r_M	Elastic follow-up factor, $r_M = T/t - 1$ in the practical RCC-MRx form (or $r_M = \frac{2(1+\nu)}{3}(T/t - 1)$ in the rigorous isotropic derivation reproduced in Appendix B).
g	Abatement coefficient applied to the displacement-controlled moment, given by Eq. (17) without pressure and by Eq. (18) when a primary membrane pressure stress σ_P is present.
σ_{ref}	Elastic reference equivalent stress used to build T and t ; for piping sections it is assembled from the weighted primary and secondary moment terms in Eq. (16).

Table 4 is intended as a quick-reference summary; its notation is used consistently throughout the remainder of the paper and in Appendix B.

This raises a natural verification question, addressed by the benchmark of this paper: *is it necessary to explicitly reconsider secondary loads in the primary-stress validation, and if so, does the RCC-MRx analytical route (r_M , g) correctly and conservatively capture the elasto-plastic reality?*

7 Benchmark description

7.1 Geometry, material and boundary conditions

The benchmark model consists of a single elbow connecting two straight pipe runs, with fixed anchors at both ends, as shown in Figure 22:

- Cross-section: outer diameter 219.07 mm, wall thickness 3.76 mm;
- Straight lengths (nodes 10–20 and 30–40): 2000 mm each;
- Elbow bend radius: 500 mm;
- Fixed anchors at end nodes 10 and 40;
- Thermal expansion loading from 20 °C to 550 °C;
- No internal pressure;
- Material properties at 550 °C: $E = 155$ GPa, $\nu = 0.3$, $\alpha = 18.5 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$, $R_{p0.2} \approx 106$ MPa, Ramberg–Osgood parameters $n_0 = 0.1125$, $C_0 = 1.198$.

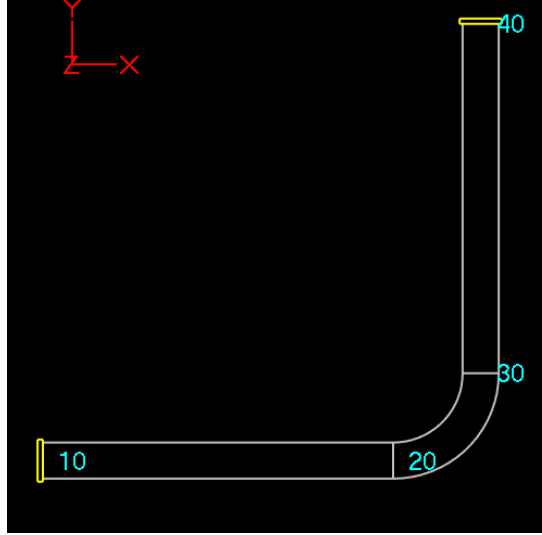


Figure 22: Benchmark geometry: elbow connecting two straight pipe runs, with fixed anchors at end nodes 10 and 40, and the elbow tangent points at nodes 20 and 30.

7.2 Numerical models

Two independent numerical models are used and cross-checked:

- **Aspect Nuclear Pipe Stress** (“NS” solver, PEPS 8.0), a beam-element piping-code model in which the elbow flexibility is represented through the analytical flexibility factor k of Eq. (4);
- **MSC Marc**, a shell finite-element model using bilinear thick-shell elements (four-node, Marc Element 75), in which elbow ovalisation is captured directly by the shell discretisation rather than through an analytical flexibility factor. This element uses bilinear interpolation of coordinates, displacements and rotations, with membrane strains from the displacement field and curvatures from the rotation field; transverse shear strains are evaluated at the mid-edges and interpolated to the Gauss integration points, giving an efficient element that remains accurate in the thin-shell limit while avoiding tying constraints at plate intersections.

The shell-element kinematics are illustrated schematically in Figure 23; for completeness, Figure 24 reproduces the software-manual extract cited in the presentation to justify this element choice for a geometrically non-linear, shell-based elbow benchmark.

The two ends of the model are connected to the FE mesh through RBE2 rigid-body elements, as shown in Figure 25. RBE2 links can locally generate spurious over-stiffening and stress concentrations; this is considered acceptable here because stresses are not examined at the anchor points themselves (only the elbow region is of interest for stress results), so that the simpler RBE2 connection is preferred over an RBE3 connection.

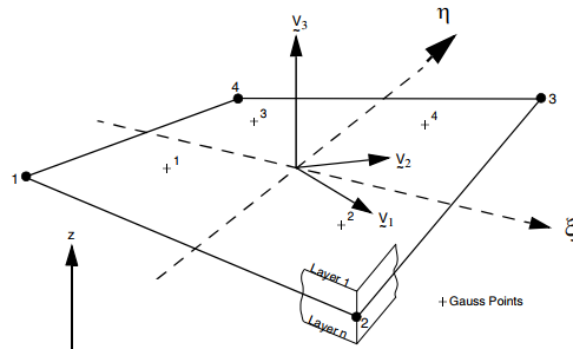


Figure 23: Schematic of the four-node bilinear thick-shell element (Marc Element 75) used for the MSC Marc model: corner nodes 1–4, local director v_3 and in-plane axes v_1 , v_2 , curvilinear coordinates ξ , η , and through-thickness Gauss integration points across layers.

Element 75

Bilinear Thick-shell Element

This is a four-node, thick-shell element with global displacements and rotations as degrees of freedom. Bilinear interpolation is used for the coordinates, displacements and the rotations. The membrane strains are obtained from the displacement field; the curvatures from the rotation field. The transverse shear strains are calculated at the middle of the edges and interpolated to the integration points. In this way, a very efficient and simple element is obtained which exhibits correct behavior in the limiting case of thin shells. The element can be used in curved shell analysis as well as in the analysis of complicated plate structures. For the latter case, the element is easy to use since connections between intersecting plates can be modeled without tying.

Due to its simple formulation when compared to the standard higher order shell elements, it is less expensive and, therefore, very attractive in nonlinear analysis. The element is not very sensitive to distortion, particularly if the corner nodes lie in the same plane. All constitutive relations can be used with this element.

Figure 24: Documentation excerpt reproduced from the source presentation for Marc Element 75, emphasising why this element was selected for the benchmark: bilinear thick-shell kinematics, efficient treatment of transverse shear, acceptable behaviour in the thin-shell limit, and suitability for non-linear shell analyses.

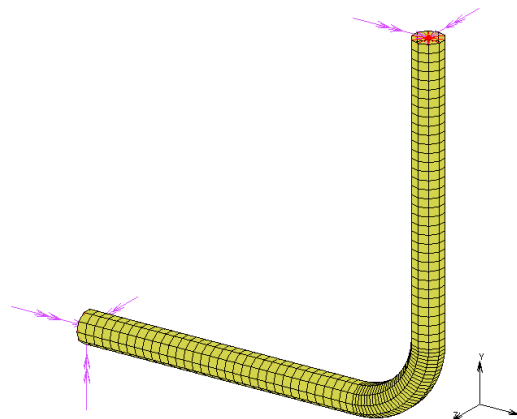


Figure 25: Detail of the shell-model anchor connection used in MSC Marc: the pipe ends are tied to the master nodes through RBE2 spiders, so that the prescribed end restraints act on rigid end planes rather than on individual shell nodes.

8 Elastic analysis and model calibration

Before introducing plasticity, the two models are calibrated against one another on a purely elastic basis. Because the beam model represents elbow ovalisation only through the analytical flexibility factor k , whereas the shell model captures it directly through the finite-element discretisation, the anchor reaction moment (at the RBE2 master node) is used as the key comparison quantity, since it is highly sensitive to the overall flexibility of the system.

Table 5 reports the anchor reaction moment obtained with the beam model with and without the elbow flexibility factor, and with the shell model at increasing circumferential mesh densities.

Table 5: Anchor reaction moment: beam model (with/without flexibility factor) versus shell FE model at increasing mesh density.

Model	Reaction moment [N m]
NS beam model, without k -factor	56 702
NS beam model, with k -factor ($k = 10.1717$)	39 632
MSC Marc shell, 10 elements/circumference	38 260
MSC Marc shell, 20 elements/circumference	39 650
MSC Marc shell, 40 elements/circumference	40 180
MSC Marc shell, 80 elements/circumference	40 290

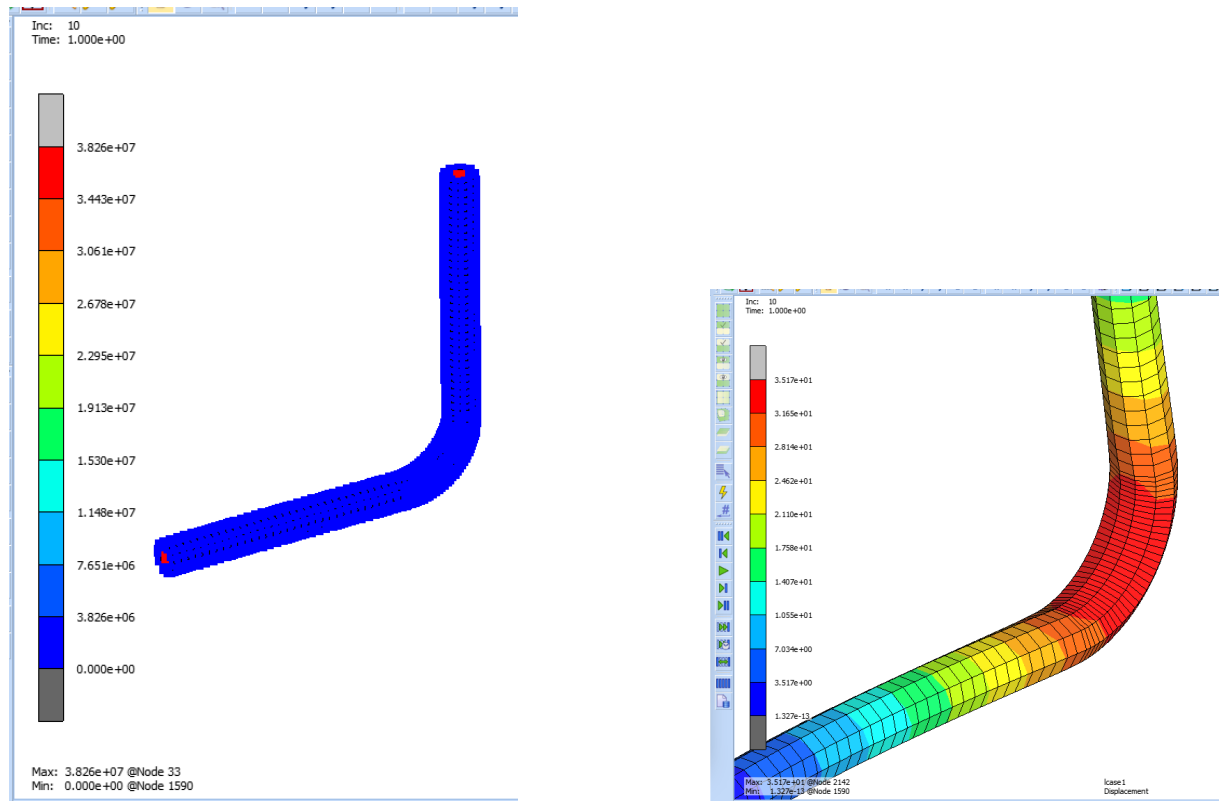


Figure 26: MSC Marc elastic shell model of the benchmark elbow. Left: overall geometry and elastic stress field, anchored at both ends. Right: deformed shape (magnified) coloured by displacement magnitude, showing the ovalisation-driven flexibility of the elbow region (maximum displacement 35.17 mm at the free end, node 2142; anchor node 1590).

The presentation also documents the same run carried through to its final increment on the finer shell mesh, reproduced in Figure 27: the maximum displacement stabilises at 35.25 mm (node

8121), essentially unchanged from the coarser/earlier-increment view of Figure 26, confirming that the elastic solution is converged with respect to load increment.

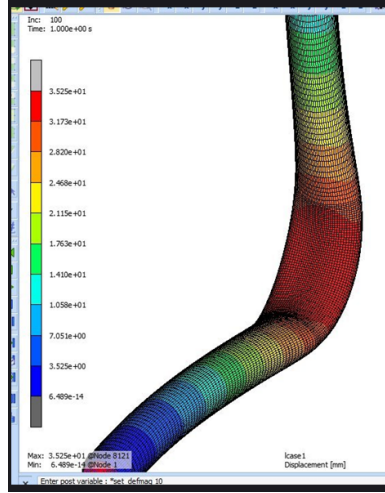


Figure 27: MSC Marc elastic shell model, final increment (Inc. 100) on the refined mesh: deformed shape coloured by displacement magnitude, maximum 35.25 mm at node 8121 – the converged counterpart of Figure 26 (right).

The beam model without the elbow k -factor overestimates the system stiffness by about 40 % relative to the converged shell solution, confirming the essential role of elbow ovalisation in the overall flexibility of the system. Once the analytical flexibility factor of Eq. (4) is introduced ($k = 10.1717$), the beam-model reaction moment (39 632 N m) matches the shell result at a moderate mesh density (20 elements per circumference, 39 650 N m) extremely well, confirming both the validity of the analytical flexibility factor and the adequacy of the beam-element representation for this geometry. Figure 26 shows the corresponding shell-model stress field and displacement pattern. As the shell mesh is further refined to 40 and 80 elements per circumference, the reaction moment increases moderately further (to 40 180 N m and 40 290 N m, respectively), indicating a degree of mesh sensitivity in the shell model that is revisited in Section 10. Large-deformation effects (ovalisation) are included in the shell analysis.

The presentation also compares the global deformed shapes produced by the two models. The beam-based NS model predicts a maximum thermal-expansion displacement of 33.649 mm at point Z010, reported in Figure 28, which is consistent with the shell-model maximum displacement reported in Figure 26; this provides a second, kinematic confirmation that the calibrated beam model reproduces the shell-model flexibility with acceptable accuracy for this benchmark. Figure 29 further documents the shell mesh that was finally retained for the non-linear run.

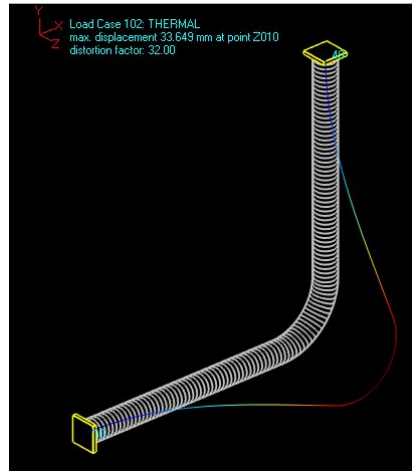


Figure 28: Deformed shape from the Aspect Nuclear Pipe Stress elastic benchmark (load case 102, thermal expansion): the beam-model maximum displacement is 33.649 mm at point Z010, close to the shell-model prediction after calibration of the elbow flexibility.

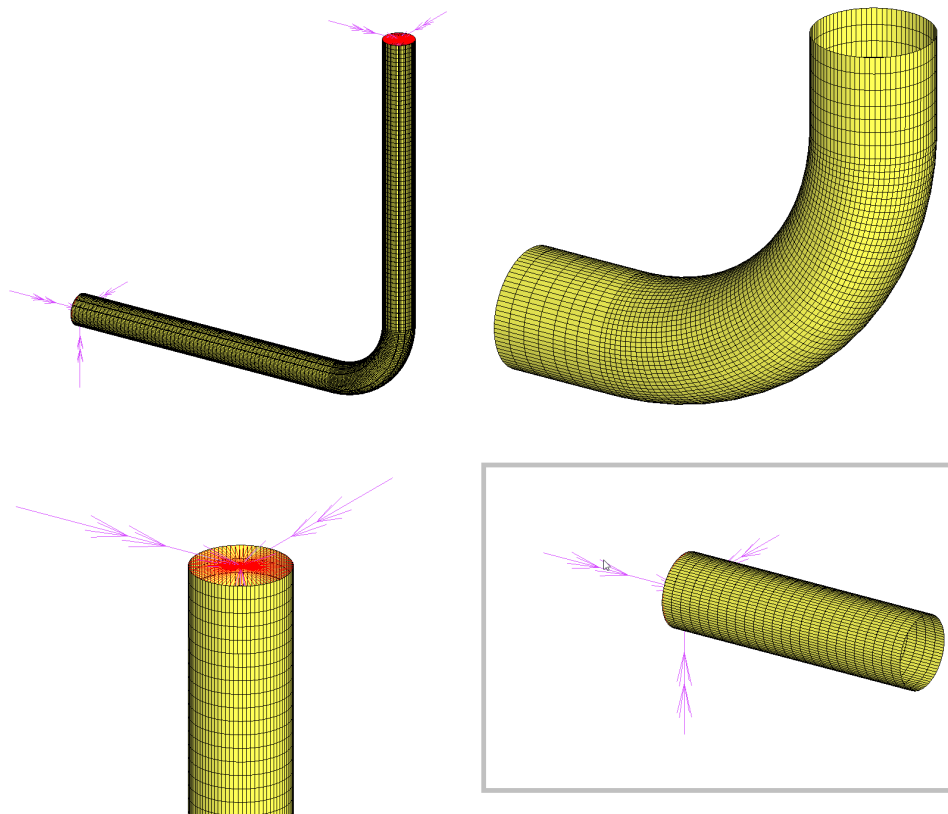


Figure 29: Additional shell-model views extracted from the presentation for the 80-elements-per-circumference discretisation used in the elasto-plastic benchmark. These views document the mesh density retained around the elbow and the associated verification plots used before running the non-linear calculation.

9 Elasto-plastic analysis

9.1 Constitutive model

The elasto-plastic shell analysis uses the von Mises yield criterion with isotropic hardening, and the Ramberg–Osgood law of Eq. (8) calibrated to 316L(N) at 550 °C (RCC-MRx RB A3.3S.451), with yielding onset defined at zero offset stress consistent with the code’s convention. The material is defined in MSC Marc by $E = 155\,000$ MPa, $\nu = 0.3$, and the von Mises yield condition $\sigma_{VM} < \sigma_Y$ with

$$\sigma_{VM} := \sqrt{\frac{3}{2} s_{ij} s_{ij}}, \quad (20)$$

s_{ij} the deviatoric stress. Hardening is tracked through the equivalent plastic strain

$$\varepsilon_{p,eq} := \sqrt{\frac{2}{3} \varepsilon_{ij}^p \varepsilon_{ij}^p}, \quad (21)$$

with ε_{ij}^p the plastic strain tensor, consistent with the standard von Mises formulation. Isotropic hardening is imposed as a tabulated function $\sigma_Y(\varepsilon_{p,eq}) := F(V1)$, with $V1$ the independent (equivalent plastic strain) variable digitised from the Ramberg–Osgood law of Eq. (8) at zero offset stress: the presentation’s *plas_new* table samples this curve over 34 points, $V1 \in [0, 1.002]$, $F \in [0.6, 213.2]$ MPa (dimensionless scale factor $\times 100$), reproduced in Figure 30.

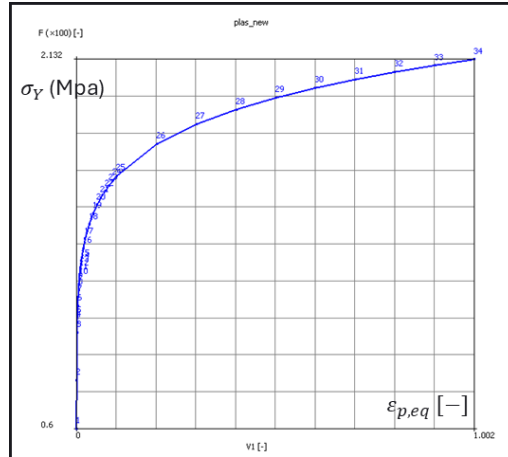


Figure 30: MSC Marc tabulated isotropic hardening curve (*plas_new* table), σ_Y versus equivalent plastic strain $\varepsilon_{p,eq}$, digitised at 34 points from the zero-offset Ramberg–Osgood law of Eq. (8) and used directly as the FE material input.

The shell mesh used for the elasto-plastic run (80 elements around the circumference) was checked beforehand for consistency with the elastic reference solution of Section 7. The source presentation makes this consistency check explicit by reading the retained critical-point values back onto the FE hardening law: the final shell result combines $\sigma_{VM} = 86.5$ MPa with an equivalent plastic strain of about $\varepsilon_{p,eq} = 0.05\%$ (Eq. (21)), and the corresponding point lies on the imposed work-hardening curve $\sigma_Y(\varepsilon_{p,eq})$ used in the Marc material definition. Figure 31 zooms into the small-strain region of this same curve (equivalent plastic strain up to 0.12%) and marks this exact point, directly confirming graphically that the FE critical-section result is consistent with the imposed hardening law.

9.2 Stress distribution and identification of the critical section

The von Mises stress distribution over the elbow shows a clearly identifiable “weak link” region (the section of maximum bending combined with ovalisation-amplified local stress), consistently located at the same position across all models, which is used as the reference cross-section for the

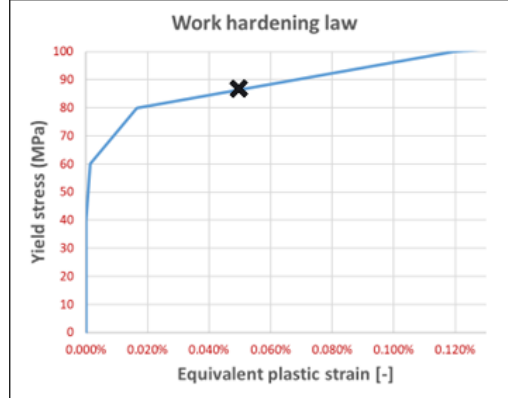


Figure 31: Zoom of the work-hardening law over the small-strain range relevant to the benchmark, with the retained critical-section point ($\sigma_{VM} = 86.5$ MPa, $\varepsilon_{p,eq} = 0.05$ %) marked, confirming its consistency with the imposed FE hardening curve.

abatement-coefficient comparison below. Figure 32 gives the full-field response, while Figure 33 zooms in on the section from which the FE-based value of g is back-calculated.

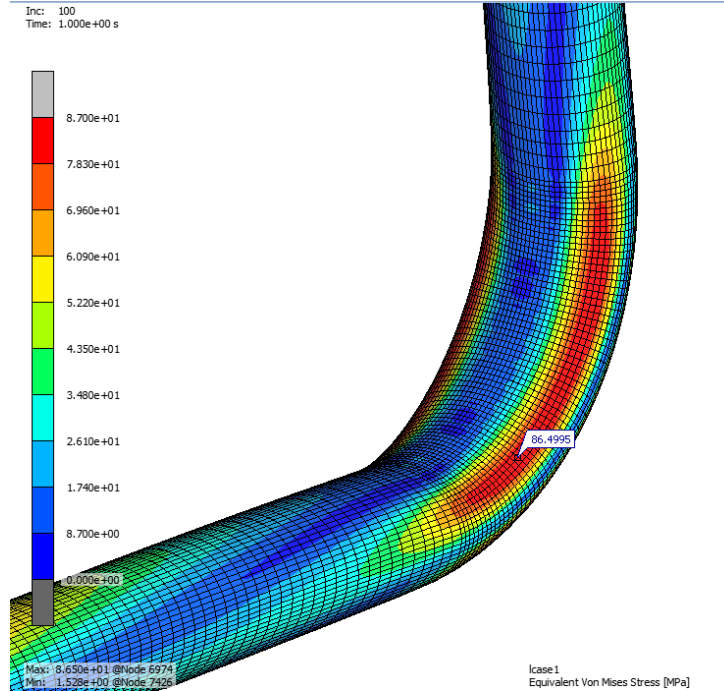


Figure 32: MSC Marc elasto-plastic shell model (80 elements/circumference): equivalent von Mises stress distribution at the end of the thermal transient, showing the critical section at the elbow crown (maximum 86.5 MPa, node 6974; minimum 1.5 MPa, node 7426).

9.3 Comparing the FE elasto-plastic and NS elastic-plus-code-rule approaches

The RCC-MRx elastic route (RB 3651.1131) reduces the secondary (displacement-dependent) moment m by the coefficient g and combines the abated stress with the primary stress in the $\sigma_m + \sigma_b$ (P-type) damage check; equivalently, g can be interpreted either as an abatement applied to the stresses (from the elastic/purely-secondary value down to the effective value) or as an abatement applied to the moments (from the elastic moment down to the effective moment). The natural benchmark question is then: *how does the RCC-MRx-derived g (and the associated*

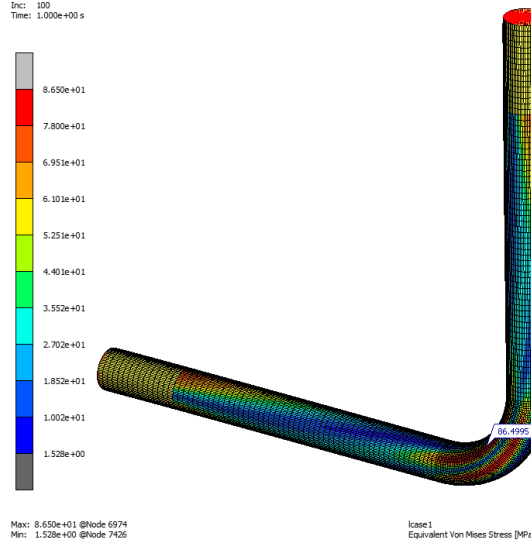


Figure 33: Close-up of the critical elbow section used for the FE/RCC-MRx comparison: the local maximum equivalent von Mises stress of 86.5 MPa is the reference value from which the FE-based abatement coefficient is back-calculated.

$\sigma_m + \sigma_b$) compare with the coefficient and stress obtained from a full elasto-plastic FE reference calculation at the same critical section?

Table 6 reports this comparison.

Table 6: Abatement coefficient g and equivalent stress: RCC-MRx analytical route versus elasto-plastic FE reference.

Model / route	g	Stress [MPa]
Aspect Nuclear Pipe Stress, RCC-MRx (SMC1), $\sigma_m + \sigma_b$	0.323	66.5
MSC Marc elasto-plastic shell, reaction-based effective abatement	0.436	86.5

Taken together, Figures 34 and 35 provide the production-code evidence behind the benchmark values reported in Table 6: the first figure documents the local r_M/g calculation, while the second gives the corresponding stress result at the critical point. The source presentation additionally distinguishes two FE-side ways of defining an “effective” abatement coefficient from the shell model: a local stress ratio, $g_{VM} = 86.5/213.7 = 0.405$, and a reaction-moment ratio, $g_{reaction} = 17.58/40.3 = 0.436$. The presentation retains the second value in the comparison table because RCC-MRx applies g to the displacement-dependent moment rather than directly to the local von Mises stress.

Two observations follow directly from Table 6:

1. **Observation 1:** the RCC-MRx abatement coefficient ($g = 0.323$) is less conservative than the coefficient implied by the elasto-plastic FE reference ($g = 0.436$).
2. **Observation 2:** the RCC-MRx primary-equivalent stress ($\sigma_m + \sigma_b = 66.5$ MPa) is less conservative than the von Mises stress of the reference elasto-plastic FE analysis (86.5 MPa).

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The following table summarizes the calculations of monotonic elastic follow-up (max value rM) and abatement coefficient g:

Element	LOAD CASE NO. 202		EpsMpRef1	EpsMpRef2	Max(1/t)	Sigma	SigmaP	EpsMpSig	EpsMpSigP	T	rM	g
	SigmaRef1 (MPa)	SigmaRef2 (MPa)	(%)	(%)		(MPa)	(MPa)	(%)	(%)			
0	232.63	227.33	217.195	176.998	1447.18	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
1	227.33	222.04	176.998	143.546	1206.82	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
2	222.04	216.74	143.546	115.830	1002.07	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
3	216.74	211.45	115.830	92.970	828.34	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
4	211.45	206.15	92.970	74.207	681.51	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
5	206.15	200.86	74.207	58.884	557.94	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
6	200.86	195.56	58.884	46.438	454.41	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
7	195.56	190.27	46.438	36.384	368.06	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
8	190.27	184.97	36.384	28.311	296.40	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
9	184.97	179.68	28.311	21.870	237.24	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
10	179.68	174.38	21.870	16.764	188.66	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
11	174.38	169.09	16.764	12.745	149.01	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
12	169.09	163.79	12.745	9.606	116.83	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
13	163.79	158.50	9.606	7.173	90.90	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
14	158.50	153.20	7.173	5.303	70.14	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
15	153.20	147.91	5.303	3.879	53.65	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
16	147.91	142.61	3.879	2.805	40.65	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
17	142.61	137.32	2.805	2.004	30.49	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
18	137.32	132.02	2.004	1.413	22.62	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
19	132.02	126.73	1.413	0.982	16.59	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
20	126.73	121.44	0.982	0.672	12.01	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
21	121.44	116.14	0.672	0.452	8.58	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
22	116.14	110.85	0.452	0.299	6.03	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
23	110.85	105.55	0.299	0.193	4.18	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
24	105.55	100.26	0.193	0.122	2.84	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
25	100.26	94.96	0.122	0.076	1.89	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
26	94.96	89.67	0.076	0.045	1.23	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
27	89.67	84.37	0.045	0.026	0.78	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
28	84.37	79.08	0.026	0.015	0.49	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
29	79.08	73.78	0.015	0.008	0.29	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
30	73.78	68.49	0.008	0.004	0.17	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
31	68.49	63.19	0.004	0.002	0.09	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
32	63.19	57.90	0.002	0.001	0.05	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
33	57.90	52.60	0.001	0.000	0.02	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
34	52.60	47.31	0.000	0.000	0.01	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
35	47.31	42.01	0.000	0.000	0.01	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
36	42.01	36.72	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
37	36.72	31.42	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
38	31.42	26.13	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
39	26.13	20.83	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
40	20.83	15.54	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
41	15.54	10.24	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237
42	10.24	4.95	0.000	0.000	0.00	119.59	0.00	0.587	0.000	0.0010	2.638	0.3237

Figure 34: Actual production-code output (Aspect Nuclear Pipe Stress / HEXAGON NEW SOLVER, PC-ISL-64b) for the benchmark elbow under thermal load case LC 202: the element-by-element elastic follow-up calculation gives $r_M = 2.638$ and $g = 0.3237$ at the critical section, confirming the value reported in Table 6.

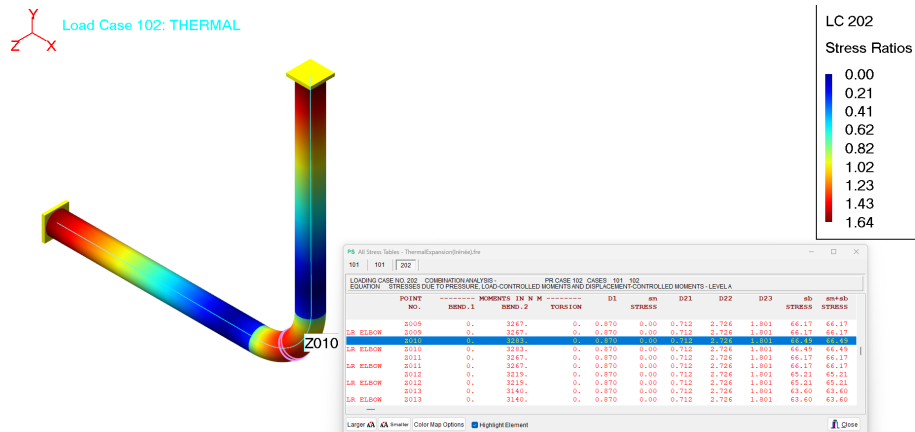


Figure 35: Aspect Nuclear Pipe Stress (NS solver) results for the thermal expansion load case (LC 202): stress-ratio colour map along the model and detailed stress table at the elbow points (Z009–Z013), giving $\sigma_m + \sigma_b = 66.49$ MPa at the critical point Z010, consistent with Table 6.

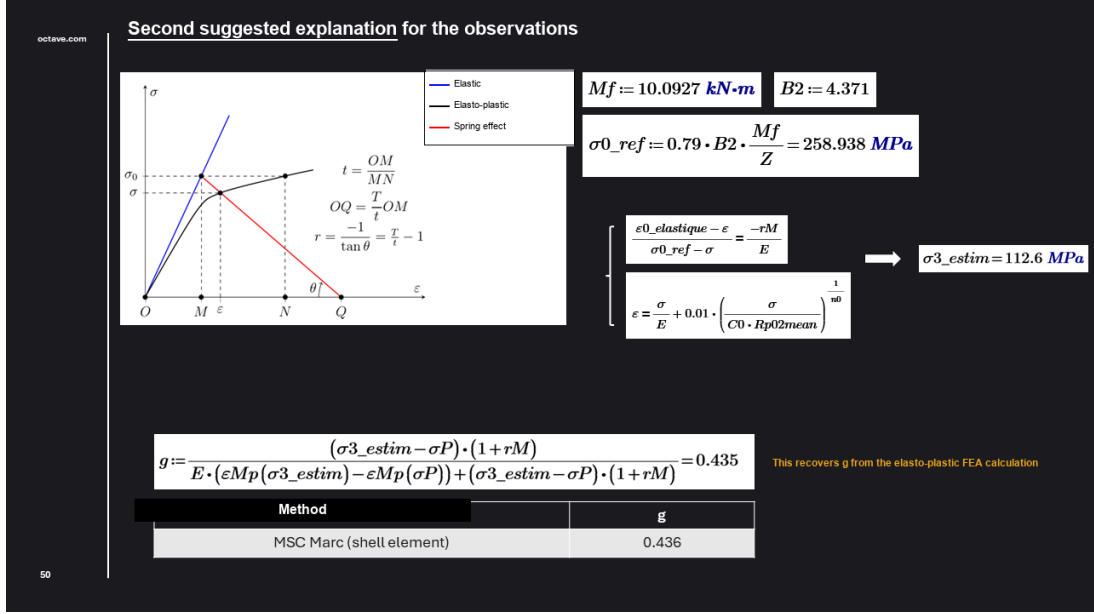


Figure 36: Second FE-side consistency check reproduced from the presentation: starting from the shell-model reaction moment $M_f = 10.0927 \text{ kN m}$ and the piping-code coefficient $B_2 = 4.371$, the equivalent elastic reference stress is reconstructed as $\sigma_{0,ref} = 0.79 B_2 M_f / Z = 258.938 \text{ MPa}$. Solving the coupled elastic-follow-up and Ramberg–Osgood relations shown on the slide then gives $\sigma_{3,est} = 112.6 \text{ MPa}$ and $g = 0.435$, i.e. the reaction-based FE abatement coefficient $g_{reaction} = 0.436$ to within rounding.

10 Discussion: why is the RCC-MRx route less conservative?

10.1 First explanation: primary-equivalent stress conversion

Slide 44 of the source presentation first recalls clause RB 3651.111, titled *Limitations des contraintes dues à la pression*, which gathers the classical primary-pressure limits

$$P_m \leq S_m(\theta_m), \quad P_L \leq 1.5 S_m(\theta_m), \quad P_L + P_b \leq 1.5 S_m(\theta_m).$$

Within that same elastic-analysis framework, the subsequent conversion argument then moves to RB 3651.112, i.e. the general primary membrane-plus-bending check used here. RCC-MRx clause RB 3651.112 requires that, for load combinations governed by level-A criteria, the general primary membrane-plus-bending stress additionally satisfies

$$\overline{p_m + p_b} = B_1 P \left[\frac{D_e}{2h_c} \right] + \langle B_2/Z, M \rangle \leq 1.3 S_m(\theta_m), \quad (22)$$

i.e. the abated, displacement-dependent stress must ultimately be re-expressed and re-checked on the strictly primary allowable basis $1.3 S_m(\theta_m)$, rather than on the CMS1 allowable S^* of Eq. (19) used to derive g in Section 6. For the benchmark section considered here, $1.3 S_m = 100.1 \text{ MPa}$ while $S^* = 48.156 \text{ MPa}$, so that the ratio

$$\frac{1.3 S_m}{S^*} = \frac{100.1}{48.156} = 2.079 \quad (23)$$

is the conversion factor that must be applied when re-scaling the RCC-MRx abatement coefficient and stress from the S^* (CMS1) basis onto the $1.3 S_m$ (primary) basis:

$$g_{\text{RCC-MRx}}^{\text{rabat}} = g \cdot \frac{1.3 S_m}{S^*} = 0.323 \times 2.079 = 0.671, \quad (24)$$

$$(\sigma_m + \sigma_b)^{\text{rabat}} = (\sigma_m + \sigma_b) \cdot \frac{1.3 S_m}{S^*} = 138.23 \text{ MPa}.$$

Once this conversion is applied, the RCC-MRx-based intensity of primary-type stress ($g^{\text{rabat}} = 0.671$, $(\sigma_m + \sigma_b)^{\text{rabat}} = 138.23 \text{ MPa}$) becomes markedly *more* conservative than the elasto-plastic FE reference ($g = 0.436$, $\sigma_{\text{VM}} = 86.5 \text{ MPa}$), reversing the apparent non-conservatism identified in Table 6. This confirms that the raw comparison of Table 6 is not made on a consistent allowable-stress basis: g and $\sigma_m + \sigma_b$ as computed under the CMS1 rule (RB 3651.1131, allowable S^*) are not directly comparable to a von Mises stress checked against $1.3 S_m$, and the code-mandated re-scaling of Eq. (22) restores the expected conservatism of the analytical RCC-MRx route.

10.2 Second explanation: mesh sensitivity of the reference solution

A complementary explanation emerges from the mesh sensitivity already noted in Table 5. Two distinct pieces of evidence are available in the source presentation. First, slide 38 contains a native Office chart titled *Mesh convergence analysis*, whose exact data are reproduced in Appendix A (Table 7). Those four points are the maximum von Mises stresses extracted from the shell calculations, not the g values themselves: the raw stress sequence is 105.02 MPa, 83.86 MPa, 84.57 MPa and 86.5 MPa for 10, 20, 40 and 80 circumferential elements, respectively, so it is not strictly monotonic. The key observation is that the coarse 10-element mesh is an outlier, whereas the 20-, 40- and 80-element meshes cluster within a much narrower band and approach the retained elasto-plastic reference value of 86.5 MPa. Second, the separate NS/NewSolver sensitivity plot reproduced in Figure 37 tracks the derived RCC-MRx abatement coefficient as the discretisation is refined; it shows an overall upward stabilisation trend with small oscillations, rather than a perfectly monotonic increase. Taken together, these two slides indicate that once mesh convergence is properly achieved, the discrepancy between the RCC-MRx analytical route and the elasto-plastic reference is substantially reduced, even before the primary-equivalent stress conversion of Section 10 is applied. This suggests that part of the discrepancy originally observed in Table 6 originates from an insufficiently refined shell mesh in the reference elasto-plastic model, rather than from an intrinsic shortcoming of the RCC-MRx r_M/g formulation itself.

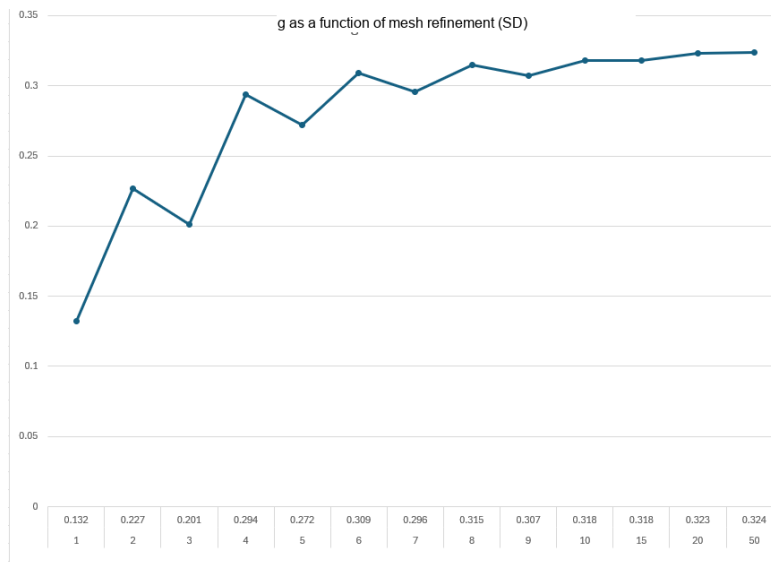


Figure 37: NS/NewSolver mesh-sensitivity plot extracted from the source presentation (slide 54), showing the derived RCC-MRx abatement coefficient g as a function of discretisation refinement. The curve exhibits an overall upward convergence trend with small oscillations, complementing the separate shell-model stress data reported in Appendix A, Table 7.

Both explanations are consistent with one another and are not mutually exclusive: the primary-equivalent stress conversion is a code-mandated, conservatism-restoring step that must always be applied; the mesh-sensitivity finding is a numerical-modelling caveat that must be

controlled independently, regardless of which analytical route (RCC-MRx or direct elasto-plastic FE) is used.

10.3 Additional validation: comparison across wall-thickness variants

As a further, independent check, the RCC-MRx abatement coefficient g obtained from the elasto-plastic FE reference (g_{FEA}) was compared against the RCC-MRx analytical value ($g_{\text{RCC-MRx}}$) on two additional model variants of the same benchmark, differing only in wall thickness (12 mm and 15 mm), including large-strain and plasticity effects. The comparison is summarised in Figure 38. In both cases, the RCC-MRx value bounds the FE-derived value, confirming the conservatism of the code-mandated coefficient across a range of section stiffnesses.

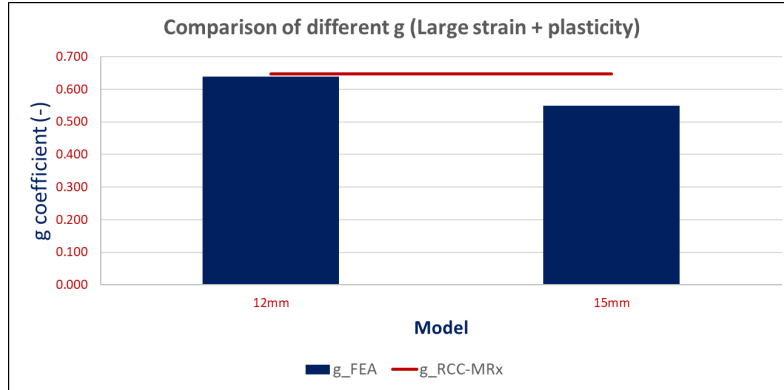


Figure 38: Comparison of the abatement coefficient g from a large-strain elasto-plastic FE calculation (g_{FEA}) against the RCC-MRx analytical value ($g_{\text{RCC-MRx}}$), for two wall-thickness variants (12 mm and 15 mm) of the benchmark elbow. The RCC-MRx value remains conservative (equal to or above g_{FEA}) in both cases.

10.4 Analytical validation on a cantilever with imposed end displacement

To validate the production-code implementation of r_M and g independently of the elbow benchmark, a simple, closed-form case is considered: a straight cantilever beam of length l , clamped at one end, subjected to an imposed transverse displacement Δu at the free end (Figure 39). In the benchmark slide deck, the imposed displacement is specified explicitly as $\Delta u = 12$ mm. In the closed-form derivation below, the curvilinear abscissa x is measured from the free end ($x = 0$) towards the clamp ($x = l$), so that the elastic bending moment and the elastic reference stress vary linearly with x .

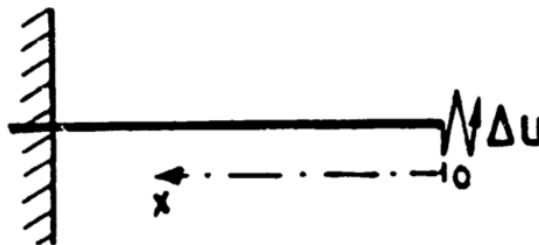


Figure 39: Schematic of the cantilever validation case: a beam of length l , clamped at one end and subjected to an imposed transverse displacement Δu at the other; for the analytical derivation, x is measured from the free end to the clamped end.

For this configuration, the elastic reference stress is linear in x , $\sigma_{\text{ref}}(x) = \omega x$ with $\omega =$

$0.79 (3EI/l^3) \Delta u/Z$, and the local resilience $1/t(x)$ follows directly from the Ramberg–Osgood plastic strain $\varepsilon_{M,p}$. Carrying out the integrals of Eq. (16) in closed form for this linear stress distribution gives the elastic follow-up factor as an explicit function of position,

$$r(x) = \frac{T}{t} - 1 = \frac{1 + 2n_0}{3n_0} \left(\frac{x}{l} \right)^{\frac{1-n_0}{n_0}} - 1, \quad (25)$$

which is monotonically increasing towards the clamp and is therefore maximised at the clamped end ($x = l$), giving the closed-form bound

$$r_{\max} = \frac{1 - n_0}{3n_0}. \quad (26)$$

For the benchmark material ($n_0 = 0.1125$), the same closed-form development gives the global resilience quantity $T = 0.444$, and Eqs. (25)–(26) then give $r_{\max} = 2.63$. Using the no-pressure RCC-MRx expression for the abatement coefficient on that analytical solution yields $g = 0.3118$, i.e. essentially the same value as the production-code result $g = 0.3116$ reported below. Figures 40, 41 and 42 show, respectively, the FE model, the non-linear stress field and the production-code output used to validate this closed-form result.

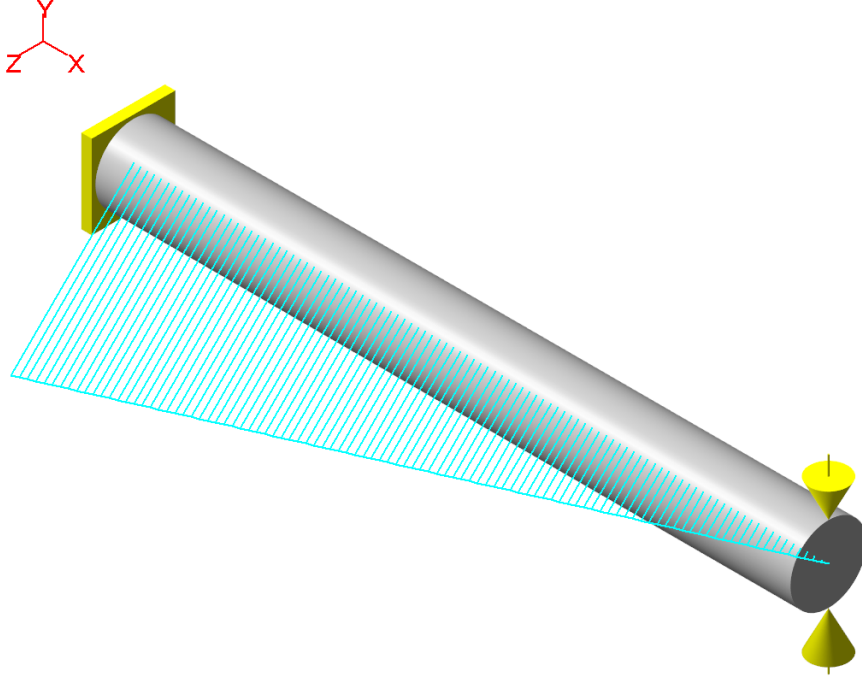


Figure 40: Finite-element model of the cantilever validation case: clamped at one end (yellow anchor), imposed displacement enforced through a rigid link at the free end (yellow cones).

The close agreement between the closed-form analytical bound $r_{\max} = 2.63$ and the production-code result ($r_M = 2.627$) constitutes an independent, purely analytical validation of the code’s implementation of the RCC-MRx elastic follow-up factor and abatement coefficient, on a case entirely decoupled from the elbow ovalisation and shell-modelling effects discussed above.

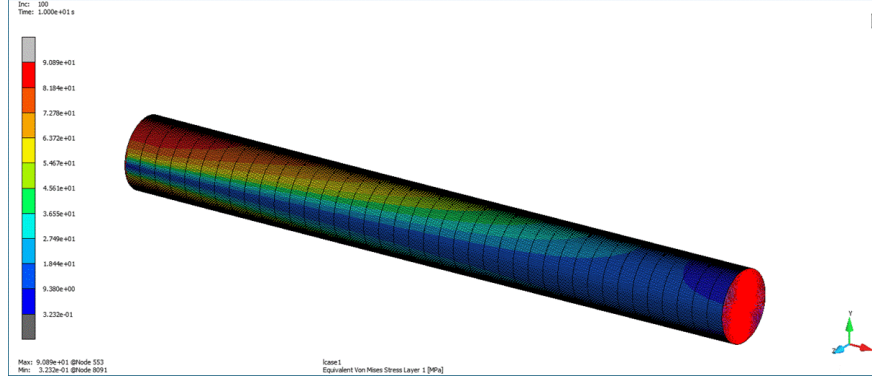


Figure 41: Elasto-plastic MSC Marc result for the simplified validation benchmark used to check the r_M/g implementation: the equivalent von Mises field confirms the stress localisation pattern that is then compared against the elastic-plus-RCC-MRx route.

□ H E X A G O N K-7.1 PAGE NO. 17

++ HEXAGON/NEW SOLVER ++ RCC-MRx type P Version 01.00.00.0295 PC-ISL-64b Release: Jul 2025

The following table summarizes the calculations of monotonic elastic follow-up (max value r_M) and abatement coefficient g :

Element	LOAD CASE NO. 501		EpsMpRef1	EpsMpRef2	Max(l/t)	Sigma	SigmaP	EpsMpSig	EpsMpSigP	T	r_M	g
	SigmaRef1	SigmaRef2	(%)	(%)		(MPa)	(MPa)	(%)	(%)			
0	118.04	116.86	0.523	0.478	6.86	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
1	116.86	115.68	0.478	0.437	6.34	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
2	115.68	114.50	0.437	0.399	5.85	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
3	114.50	113.32	0.399	0.363	5.40	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116
4	113.32	112.14	0.363	0.331	4.97	120.39	0.00	0.622	0.000	0.5286	2.627	0.3116

Figure 42: Production-code output (Aspect Nuclear Pipe Stress / HEXAGON NEW SOLVER, RCC-MRx type P) for the cantilever validation case (LC 501): the code reports $r_M = 2.627$ at the critical section, in close agreement with the closed-form bound $r_{\max} = 2.63$ of Eq. (26), together with $g = 0.3116$.

11 Conclusions

1. The paper provides a consolidated route from the historical background and constitutive assumptions to the practical RCC-MRx definitions of r_M and g , supported by the original diagrams, code excerpts and derivation material from the source presentation.
2. The elastic benchmark shows that the beam-element piping model and the MSC Marc shell model agree well in terms of global flexibility once the elbow flexibility factor is introduced consistently and the boundary conditions are treated carefully.
3. For the benchmark elbow, the direct RCC-MRx output gives $r_M = 2.638$ and $g = 0.3237$, whereas the non-linear shell reference yields a higher effective abatement level when the comparison is made on a raw section-stress basis.
4. This apparent discrepancy is resolved by two complementary observations: the code requires the abated secondary effect to be re-expressed on a primary-stress basis, and the shell reference solution is itself sensitive to mesh density in the elbow ovalisation zone (see also Appendix A).
5. The additional wall-thickness variants and the cantilever validation case support the robustness of the implementation and show that the analytical elastic follow-up route can be checked independently of the main elbow benchmark.
6. Modern numerical tools make such comparisons much faster to perform, but they also increase the importance of modelling choices and verification steps, especially those related to mesh density, shell formulation and boundary-condition idealisation.
7. The present work remains limited by the adopted boundary conditions, shell formulation, temperature-invariant Ramberg–Osgood shape parameters, and the absence of creep, relaxation and pressure effects. Extending the study to include pressure loading, alternative hardening descriptions and further mesh-convergence work is a natural next step.

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A Sensitivity of the RCC-MRx abatement coefficient to mesh density

As introduced in Section 10, the source PowerPoint contains one native embedded chart object (slide 38, chart title *Mesh convergence analysis*) giving the exact shell-model maximum von Mises stresses listed in Table 7. This native chart is distinct from Figure 37, which comes from slide 54 and plots the *derived* RCC-MRx abatement coefficient g obtained in the NS/NewSolver study. The slide 38 stress data show that the raw shell stress does *not* vary monotonically with refinement: it drops from 105.02 MPa at 10 elements to 83.86 MPa at 20 elements, then rises slightly to 84.57 MPa at 40 elements and 86.5 MPa at 80 elements. The correct interpretation is therefore that the coarsest mesh is non-converged, whereas the finer meshes stabilise in the 83.86 MPa to 86.5 MPa range and recover the retained elasto-plastic reference at 80 elements. The

abatement coefficient g discussed in Section 10 is a post-processed quantity derived from these mesh-dependent shell results together with the RCC-MRx/NS calculation chain; its convergence need not mirror the raw stress points point-for-point, nor be perfectly monotonic itself.

Table 7: Exact source data of the native PowerPoint mesh-convergence chart (*Mesh convergence analysis*, slide 38): maximum shell-model von Mises stress versus circumferential mesh density.

Circumferential elements	Maximum von Mises stress [MPa]
10	105.02
20	83.86
40	84.57
80	86.50

B Roland Roche’s demonstration

This appendix reproduces, for completeness, the mathematical derivation of the elastic follow-up factor r_M from the energy balance of a purely elastic finite-element analysis, following R. Roche’s original method [4] as re-derived, in full native LaTeX (no scanned material), from an internal technical note accompanying the source presentation [5]. This is the rigorous justification of Eqs. (15)–(16) used in Section 6. Figures 43 and 44 reproduce the graphical and documentary material from which the algebraic steps below were reconstructed.

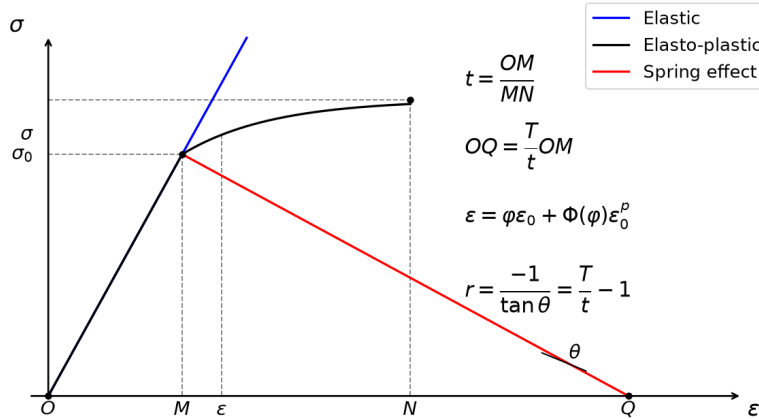


Figure 43: Graphical construction of the elastic follow-up factor r and local resilience t from the elastic (O – M – N) and elasto-plastic (O – ϵ – Q) response curves at a given imposed strain: $t = OM/MN$ and the elastic-follow-up (“spring effect”) line through (ϵ, σ) and Q has slope $-\tan \theta = -1/(T/t - 1)$.

B.1 Standard variational development in a FE analysis

The local equilibrium equation is $\sigma_{ij,j} + f_i = 0$. Admitting a variational displacement field δu applied at every material point, $\sigma_{ij,j} \delta u_i + f_i \delta u_i = 0$, and summing over the whole structure,

$$\iiint \sigma_{ij,j} \delta u_i dV + \iiint f_i \delta u_i dV = 0. \quad (27)$$

Using

$$\sigma_{ij,j} \delta u_i = (\sigma_{ij} \delta u_i)_{,j} - \sigma_{ij} \delta u_{i,j}, \quad \iiint (\sigma_{ij} \delta u_i)_{,j} dV = \iint \sigma_{ij} n_j \delta u_i dS,$$

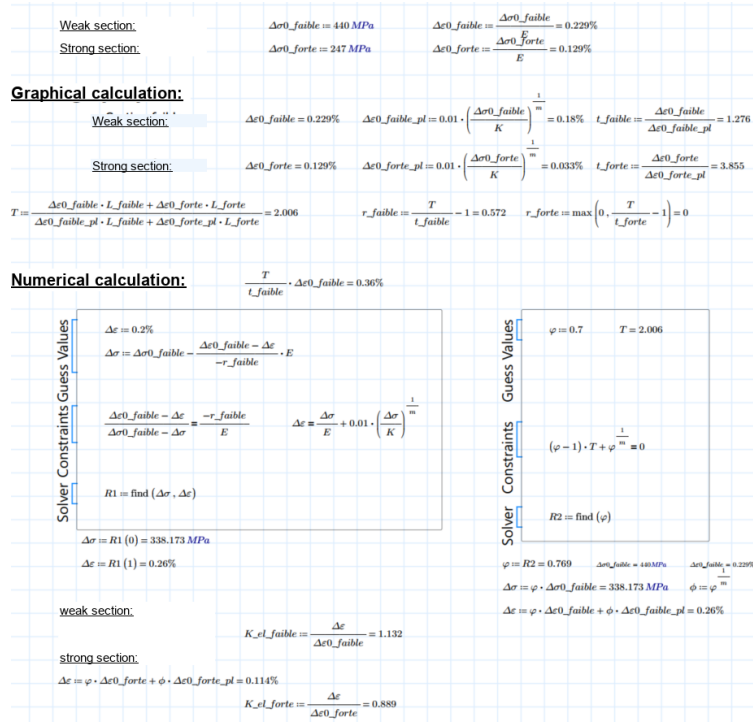


Figure 44: Numerical worksheet from the source presentation reproducing Roche's two-section example (geometry in Figure 18): the elastic reference stresses ($\Delta\sigma_{0,\text{weak}} = 440 \text{ MPa}$, $\Delta\sigma_{0,\text{strong}} = 247 \text{ MPa}$) give local resiliances $t_{\text{weak}} = 1.276$ and $t_{\text{strong}} = 3.855$, combined into a global resilience $T = 2.006$ and local elastic follow-up factors $r_{\text{weak}} = T/t_{\text{weak}} - 1 = 0.572$ and $r_{\text{strong}} = \max(0, T/t_{\text{strong}} - 1) = 0$; the lower numerical-calculation block solves the same problem with Mathcad solve blocks, confirming $\varphi = 0.769$ and $\Delta\sigma = 338.173 \text{ MPa}$ by an independent route.

the divergence theorem gives

$$\iint \sigma_{ij} n_j \delta u_i dS - \iiint \sigma_{ij} \delta u_{i,j} dV + \iiint f_i \delta u_i dV = 0. \quad (28)$$

With $\sigma_{ij} \delta u_{i,j} = \sigma_{ij} \frac{1}{2} (\delta u_{i,j} + \delta u_{j,i}) = \sigma_{ij} \delta \varepsilon_{ij}$,

$$\iint \sigma_{ij} n_j \delta u_i dS - \iiint \sigma_{ij} \delta \varepsilon_{ij} dV + \iiint f_i \delta u_i dV = 0. \quad (29)$$

Assume an external surface load

$$W_{\text{ext}} = \iint \sigma_{ij} n_j u_i dS,$$

with variational form

$$\delta W_{\text{ext}} = \iint \delta \sigma_{ij} n_j u_i dS + \iint \sigma_{ij} n_j \delta u_i dS.$$

Likewise, the internal work is

$$\delta W_{\text{int}} = \delta \iiint \bar{\sigma} : \bar{\varepsilon} dV = \iiint \bar{\sigma} : \delta \bar{\varepsilon} dV + \iiint \delta \bar{\sigma} : \bar{\varepsilon} dV.$$

The principle of virtual work, $\delta W_{\text{int}} = \delta W_{\text{ext}}$, then gives

$$\iint \delta \sigma_{ij} n_j u_i dS + \iint \sigma_{ij} n_j \delta u_i dS = \iiint \sigma_{ij} \delta \varepsilon_{ij} dV + \iiint \delta \sigma_{ij} \varepsilon_{ij} dV. \quad (30)$$

From (29)–(30), in the case $f_i = 0$, one obtains the compact equality between external complementary work and the internal stress–strain product,

$$\iint \delta \sigma_{ij} n_j u_i dS = \iiint \delta \sigma_{ij} \varepsilon_{ij} dV. \quad (31)$$

Introducing a thermal (or other imposed-strain) contribution $\bar{\bar{\varepsilon}}$, distinct from the mechanical strain $\bar{\varepsilon}$,

$$\iiint \delta \bar{\sigma} : \bar{\varepsilon} dV = \iint \delta \bar{\sigma} \bar{n} \bar{u} dS - \iiint \delta \bar{\sigma} : \bar{\bar{\varepsilon}} dV. \quad (32)$$

This is the working form used by Roche to derive r_M .

B.2 Radially proportional elasto-plastic stress field

Roche’s method assumes that the elasto-plastic stress distribution remains, at every point, homogeneously proportional to the purely elastic solution, through a single scalar φ :

$$\bar{\sigma} = \varphi \bar{\sigma}_0. \quad (33)$$

Defining the equivalent (von Mises) stress and strain through Eq. (34),

$$\varepsilon^* = \sqrt{\frac{2}{3} e_{ij} e_{ij}}, \quad e_{ij} = \varepsilon_{ij} - \frac{1}{3} \varepsilon_{kk} \delta_{ij}; \quad \sigma^* = \sqrt{\frac{3}{2} s_{ij} s_{ij}}, \quad s_{ij} = \sigma_{ij} - \frac{1}{3} \sigma_{kk} \delta_{ij}, \quad (34)$$

Eq. (33) implies $\sigma^* = \varphi \sigma_0^*$. The plastic equivalent strain follows the Ramberg–Osgood law, Eq. (8):

$$\varepsilon^*(\sigma^*) = \varepsilon_{\text{el}}^*(\sigma_0^*) + 0.01 \left(\frac{\varphi \sigma_0^*}{C_0 R_{p0.2, \text{mean}}} \right)^{1/n_0} = \varphi \varepsilon_0^* + \phi \varepsilon_0^{p*}, \quad \phi = \varphi^{1/n_0}, \quad (35)$$

with $\varepsilon_0^{p*} = 0.01 (\sigma_0^* / (C_0 R_{p0.2, \text{mean}}))^{1/n_0}$.

B.3 Energy balance

Using Hooke's law $\varepsilon_{0ij} = \frac{1+\nu}{E}\sigma_{0ij} - \frac{\nu}{E}\sigma_{0kk}\delta_{ij}$, its trace gives $\varepsilon_{0kk} = \frac{1-2\nu}{E}\sigma_{0kk}$, so that the deviatoric mechanical strain becomes

$$e_{0ij} = \varepsilon_{0ij} - \frac{1}{3}\varepsilon_{0kk}\delta_{ij} = \frac{1+\nu}{E}\sigma_{0ij} - \frac{\nu}{E}\sigma_{0kk}\delta_{ij} - \frac{1}{3}\varepsilon_{0kk}\delta_{ij} = \frac{1+\nu}{E}\sigma_{0ij} - \frac{\nu}{E}\sigma_{0kk}\delta_{ij} - \frac{1}{3}\frac{1-2\nu}{E}\sigma_{0kk}\delta_{ij}, \quad (36)$$

which simplifies, using $\sigma_{0ij} - \delta_{ij}\sigma_{0kk}/3 = s_{0ij}$, to

$$e_{0ij} = \frac{1+\nu}{E} \left(\sigma_{0ij} - \frac{1}{3}\delta_{ij}\sigma_{0kk} \right) = \frac{1+\nu}{E} s_{0ij}, \quad (37)$$

and hence

$$\varepsilon_0^* = \frac{2}{3} \frac{1+\nu}{E} \sigma_0^*. \quad (38)$$

Combining Eqs. (35) and (38) and defining the *local resilience* $t = \varepsilon_0^*/\varepsilon_0^{p*}$, algebraic manipulation gives

$$\frac{3}{2(1+\nu)} \frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*} = \frac{T}{t} - 1, \quad r_M = \frac{E(\varepsilon^* - \varepsilon_0^*)}{\sigma_0^* - \sigma^*} \Rightarrow r_M = \frac{2(1+\nu)}{3} \left(\frac{T}{t} - 1 \right), \quad (39)$$

where the *global resilience* T is obtained, by applying the energy balance of Eq. (32) to the purely elastic reference state and using Eqs. (31), (33)–(35),

$$(\varphi - 1) \iiint \sigma_0^* \varepsilon_0^* dV + \phi \iiint \sigma_0^* \varepsilon_0^{p*} dV = 0 \Rightarrow T \equiv \frac{\iiint \sigma_0^* \varepsilon_0^* dV}{\iiint \sigma_0^* \varepsilon_0^{p*} dV} = \frac{\iiint \sigma_0^{*2} dV}{\iiint \frac{\sigma_0^{*2}}{t} dV}, \quad (40)$$

which, assuming a constant Young's modulus and constant equivalent stress within each piping section, reduces to the practical, section-integrated form used by RCC-MRx, Eq. (16):

$$T = \frac{\int \sigma_0^{*2} A ds}{\int \frac{\sigma_0^{*2} A}{t} ds}. \quad (41)$$

Equation (40) is therefore the continuous-domain counterpart of the piping-code formula of Eq. (16). For steels ($\nu = 0.3$), Eq. (39) gives $r_M = 0.867(T/t - 1)$; the code's simplified, slightly more conservative form $r_M = T/t - 1$ (Eq. (15)) is obtained by taking $\frac{2(1+\nu)}{3} \approx 1$, the associated 3-D correction being reintroduced separately as a 0.87 stress index on pressure and torsional-moment terms (Eq. (16)). This completes, in native equation form, every algebraic step of the internal technical note [5] from which this appendix was reconstructed.

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